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// Europe, Bulgaria*

Part I. MAGIXTRAL

<https://magixtral.com>

I. Magixtral - Main Page

1. Sidebar

- The sidebar is located on the left side of the main page. It contains separate cells with drop-down menus and buttons.
- If you press the ">>" button, the sidebar will expand and be visible.
- If you press the "<<" button, the sidebar will collapse and be hidden.
- If you press and hold the right border of the sidebar "↔", you will be able to expand and shrink it manually.

1.1. System logo

- If you click on the cell with the operating system logo (MAGI), the "About" window will open, which contains information about it.

1.2. Insight

- If you press the toggle, the system mode will change, turning only the AGI vector layer from the System Layers cell on or off. If the switch is in the left position, the system will be in AI mode. If the switch is in the right position, the system will be in AGI mode.

1.3. Temperature

- If you change the position of the temperature slider, the temperature value of the language model will change. Through this slider, creativity can be adjusted.

1.4. Engine model – language model

- If you click on the drop-down menu with the engine model, the available language models that you can choose for use will be displayed. The base model of the system engine is "Mistral_7B_v0.3_Q6_K.gguf".

//Other language models can be added to the system, and for this purpose you need to add the name of the language model in the file ollama_list.json in the file manager by clicking on the settings button in the upper right corner, but for this purpose you need to download the model beforehand.
//At the moment, the original language model of Mistral is used. But besides it, there is another language model that is about to be uploaded to the system. The language model, which is to be uploaded, has been instruction fine-tuned by quantized low-rank adaptation (QLoRA) by using the 4-4-2 system, and this will happen after the completion of all necessary tests and additional settings.

//The fine-tuning done is on the same language model: Mistral 7B v0.3. For this purpose, a dataset (cognitive package) was used, which is a mixture of datasets (MoD – Mixture of Datasets) and contains information from the AGI layer, which is based on four pairs of algorithmic systems that are interconnected and from which three universal patterns for deductive determinism are built. Universal templates refer to the classification of interactions (for learning), structure of interactions (for understanding), and principles of interactions (for work). Each of these templates describes the universal rules for possible classifications, structures and principles when performing receiving and transmission processes

1.5. Personality

- If you click on the drop-down menu with personalities, the available personalities will be displayed.
 - If you click on a person from the menu, the language model will add the system prompt for the selected person to the prompt to generate a response.
 - If you press the "x" button next to a person, you can delete it.
 - If you press the "+" button in the cell, you can add your new personality to the system.
- //By choosing a personality, the language model begins to respond as that personality.

1.6. Emotionality

- If you click on the drop-down menu with emotions, the available emotional states (emotions) will be displayed.
 - If you click on emotionality from the menu, the language model will add the system prompt for the selected emotionality to the prompt to generate a response.
 - If you press the "x" button next to an emotion, you can delete it.
 - If you press the "+" button in the cell, you can add your new emotionality to the system.
- //By choosing emotionality, the language model begins to respond with this mood.

1.7. Value system

- If you click on the drop-down menu of the value system, the available qualities of the value system will be displayed.
 - If you press the "x" button next to some quality of the value system, you can delete it.
 - If you press the "+" button in the cell, you can add your own new quality to the system.
 - If you click the checkbox (or checkboxes) in the checkbox on the left, the quality will be marked or unmarked. You can choose multiple qualities from the value system.
- //By selecting qualities from the value system, the language model begins to apply them in its behavior in responses.
- //When creating new personalities, emotional states and qualities of a value system, it is necessary that the system prompt is short, clear and consistent in order to be successfully applied by the language model.

1.8. Documents

- If you click on the documents drop-down menu, the available documents will be displayed.
- If you press the "x" button next to a document, you can delete it.
- If you press the "import" button in the cell, you can add .docx .txt .json documents from your computer and use their context.

- If you click the checkmark (or checkboxes) in the checkbox on the left, the document will be marked or unmarked. You can select multiple documents, and the context of only those checked will be used by the language model when generating a response.

//If you press the "+" button, a window will appear with writing fields and "Cancel" and "Save" buttons, from which the window can be closed or a new .json file for personality, emotionality or value system can be saved.

1.9. System layers

- If you click on the drop-down menu with layers, you can select the vector layers to use in the system (from the checkboxes).

//Karma and other system layers cannot be selected.

//You can add vector layers through the file manager.

- If you press the "Dharma Wheel" button, a window will open where you can see information about the total number of system prompts in active mode (including those from deleted past sessions), the life cycle of the system, as well as other data.

1.10. Midextral

- If you click on the "mideXtral" cell, you will go to midextral.com.

2. System page top toolbar „1“

2.1. Tabs (portals)

- If you press the "+" button next to the tab, you can add a new tab (portal).
- If you press the "x" button in the tab, you can remove the tab (portal).
- If you press and hold the tab (portal), you can move and relocate it by drag and drop.

3. System page top toolbar „2“

- If you press the "<" button, you will go to the previous session in your history.
- If you press the ">" button, you will move on to the next session in your history.
- //These buttons can only be used in active modes, as there is no history in passive modes.
- If you press the "+ New chat" button, you will create a new session.
- If you click on the "Session bar", a drop-down menu with a list of sessions will be displayed.
- If you click on the "Search bar", you can search for sessions by name.
- If you click on a session in the drop-down menu, you will open it in the chat.

//Each session is numbered when it is created.

- If you press the "x" button per session, you can delete it.
- If you press the "rename" button of a session, you can rename it.
- If you click on the session checkboxes, you can add the context of the selected session to the current session you're working on. If you remove it, you'll remove the context from the current session.

//The context of a selected session is automatically removed when you exit the current session. This is useful for starting a new chat or for adding context from another session during an ongoing session.

- If you click on the "WEB" button, you can turn web search on or off.

//If you haven't selected a serper key for Google, the web button won't be active.

- If you press the "Serper key" button, from the buttons in the drop-down menu that will appear, you can select, add or remove the serper key.
- If you press the "Export" button, a drop-down menu will appear and you can choose to export the current session to: .pdf, .docx, .txt, .jsonl (lmft), .jsonl (ift).
//LMFT – export as a Language Model Fine-Tuning dataset;
//IFT – export as an Instruction Fine-Tuning dataset;
- If you press the "Day/Night" button, you can switch the system theme from the light ochre theme to the dark gray theme.

4. Chat workspace

- If you hover over a chat prompt, the "copy", "Edit and regen" and "Delete" buttons will appear above the prompt.
- If you press the "Copy" button, you will copy the prompt and it will be added to the clipboard of the Docstral application.
- If you press the "Edit and regen" button, the prompt will appear in the prompt input field, and in the chat, the prompt and reply below it will fade. In this situation, you have two options: In the first option, you can add the prompt in the box below, and then when you press the send or "enter" button, the faded prompt and reply will disappear and the new prompt will appear and a new reply will be generated for it in the chat. With the second option, you can delete the prompt in the box below, which refuses to perform response regeneration and disappears the fading of the prompt and response. Optional: if you want to remove the prompt completely without denying regeneration, you can click the button on the right of the "Clear text" prompt field.
- If you press the "Delete" button, you will delete the prompt and response.
- If you hover over a chat prompt, the "copy", "Edit and regen" and "Delete" buttons will appear above the prompt.
- If you press the "Copy" button, you will copy the answer and it will be added to the Docstral clipboard.
- If you press the "Edit" button, you can edit the answer, add or remove things in it, and the edited answer will be used as context from the language model, the old answer will be forgotten.
- If you press the "Cancel" button on the edit reply box, you can exit edit mode.
- If you press "Save", in the edit reply field, you can save the edited reply and exit the edit mode.
- If you press and hold in the corner of the reply edit box (in edit mode), you can change the size of the field.
- If you click the "Retry" button, you will delete the current answer and a new one will be generated in its place, using the same prompt.
- If the "Edit", "Retry", "Erase", "Regen Selected" or "Expand Selected" buttons are used, an additional "Undo" button will be displayed when you place the mouse over the answer.
- If you press the "Undo" button, you will return the response status before the change was made and a new additional "Redo" button will be displayed.
- If you press the "Redo" button, you will reverse the changed answer.

5. Chat container

- If you click on "Chat container", you can write a prompt.
- If you press the "Send" button on the right in the chat box or press "Enter", the language model will generate a response.
- If you press the "Send" button or press "Enter", the send button will turn into a "Stop" button.
- If you press the "Stop" button, you will stop generating replies and a "Stopped" sign will appear in the reply. When you enter the next prompt, the text "Stopped" will turn gray.
- If you press the "Appstrale" button on the left in the chat box, a drop-down menu will appear with the buttons to open the operating system's applications: "Webztral", "Docstral", "Imvistr", "File manager" and a link to an external site "Ask a colleague".
- If you press the "Webztral" button, you will open the internal system browser.
- If you click the "Docstral" button, you will open the internal text editor of the system.
- If you press the "Imvistr" button, you will open the internal media space of the system.
- If you click the "File manager" button, you will open the internal system file manager.
- If you click the "Ask a colleague" button, you will open "Mistral Le Chat" in your real external browser.
- If you click on "Chat container", the "Paste", "Continue" and "Explain" buttons will be displayed.
- If you click on another write field in the system, only the "Paste" button will be displayed.
- If you move the mouse away from the buttons, they will fade and eventually disappear.
- If you click on the side of the screen, the buttons will disappear.
- If you press the "Paste" button, you will paste copied text into the text box.
- If you click the "Continue" button, the language model will generate a continuation of the previous answer, using a system prompt to continue generating on the same topic.
- If you click the "Explain" button, the language model will generate an explanation of the previous answer using a system prompt to do so.

Buttons when marking (marking) text on the main page:

- If you mark text from a prompt or from a reply, buttons will appear above it.
- If you press the "Copy" button, the selected text will be copied and pasted into the Docstral clipboard.
- If you click the "Ask here" button, the selected text will be placed in the "Chat container" of the current tab (portal) with the option to add the prompt.
- If you click the "Ask elsewhere" button, the selected text will be placed in the "Chat container" in another tab (portal) with the option to add the prompt.
- If you press the "Summary" button, the language model will generate a summary of the selected text.
- If you press the "Explain" button, the language model will generate an explanation of the selected text.
- If you press the "v" button, a drop-down menu with additional buttons will appear.
- If you press the "^" button, you can hide the button drop-down menu.
- If you press the "Find inside" button, the selected text will be searched on Google in the internal browser (Webztral) that will open.
- If you press the "Find outside" button, the selected text will be searched on Google in the external browser (the real browser), which will open a new tab.

- If you press the "Mark as code" button, the selected text will be placed in a code block in the chat.
- If you mark a code in a code block, you can press the "Mark as text" button, which will extract the code from the block and paste it as text in the chat. If you extract code from the middle of a code block, two code blocks will appear and the extracted text will be in between.
- If you press the "Erase" button, the selected text will be removed from the prompt or response and forgotten by the system.
- If you press the "Link it" button, a window will appear with a writing box and the "Save" and "Cancel" buttons. If you enter a link and press "Enter" or "Save", the selected text will become a hyperlink, and if you press "Cancel" you will stop creating a link.
- If you click on "Regen selected", the selected text will disappear from the reply and will appear in the "chat container" as a prompt, then there are two options. With the first option, if you enter an addition to this prompt, the language model will generate a response and replace only the selected text. With the second option, if you delete the prompt from the writing field, it will reappear in the text and the regeneration process will be terminated.
- If you press "Expand selected", the selected text will disappear from the reply and will appear in the "chat container" as a prompt, after which there are two options. With the first option, if you enter an addendum to this prompt, the language model will generate an extended response and replace only the selected text. With the second option, if you delete the prompt from the writing field, it will reappear in the text and the expansion process will be terminated.
- If you press the "Astral projection" button, the selected text will be placed in a new cell in the "Astral projection" window.
- If you move the mouse away from the button drop-down menu, it will fade and eventually disappear. If you click on the side of the screen, the button drop-down menu will disappear.

6. Links:

- If you click on a link in the chat, a drop-down menu with buttons will appear.
- If you click the "Open inside" button, you will open the link in Webztral.
- If you click "Open outside", you will open the link in the real browser.
- If you click "Edit link", you can edit the link from the window that appears.
- If you press "Unlink", you will remove the hyperlink.

7. Session navigation bar

- If you press the "<<" button, you will be taken to the first prompt in the current session.
- If you press the "<" button, you will be taken to the previous prompt.
- If you press the ">" button, you will be taken to the next prompt.
- If you press the ">>" button, you will be taken to the last prompt.
- If you type a number in the typing box on the navigation bar and press "Enter", you will be taken to the prompt with the number entered. You can see the total number of prompts in the current session and the prompt you're currently on.

//To the right of each prompt, there is a number for the next number of the prompt in the current session.

8. Addons

- If you press the "Clear text" button, you will delete available text in the "Chat container" and the chat box will be pressed automatically, thus you can directly write a new prompt.
- If you press the "Astral projection" button, you will open the astral projection window.

9. Code block

- If the language model generates code, it will appear in the chat in the "code block".
- The generated code in the "code block" is "Syntax highlighted".
- Each "Code block" has line numbering as in the IDE.
- Each "Code block" contains information about the number of lines and the number of characters in the lower left part.
- Each "Code block" has buttons:
 - If you press the "Astral" button at the bottom of the code block, you will add the code from the block to a new cell in the "Astral projection" window.
 - If you click the "Copy" button at the top of the code block, you will copy the code from the block, but it will not be added to the Docstral clipboard.
 - If you press the "Unlock" button at the top of the code block, you will be able to edit the code manually and the button will change to "Lock".
 - If you press the "Lock" button at the top of the code block, you will save the edited code and the button will change to "Unlock".
 - If you press the "Export" button, a drop-down menu with file formats will appear. If you press a button: .txt, .json, .bat, .docx, .py, .html, .js, .css, .c, .cpp, you will download a file in this format that contains the code.

10. System page bottom toolbar „1“

- If you press the "Fully passive" button, you will activate this working mode in the open tab.
 - If you press the "Partially passive" button, you will activate this working mode in the open tab.
 - If you press the "Fully active" button, you will activate this working mode in the open tab.
 - If you press the "Partially active" button, you will activate this working mode in the open tab.
- //By changing the operating modes, the way the system works is changed.

II. Magixtral - Appstrale (applications) and addons

1. WEBZTRAL

1.1. Webztral window topbar:

- If you press the "Minimize" button, Webztral will hide and a Webztral bubble will appear on the main page.
- If you click on the Webztral bubble, Webztral will open.
- If you press the "Maximize" button, or if you double-click on the Webztral window topbar while it is in full-screen mode, you will switch to window mode.
- If you press and hold the borders of the Webztral window, you can change the size of the window.
- If you double-click on the Webztral window topbar while it is in window mode, you will switch to full-screen mode, and if it is the other way around, you will switch to window mode.
- If you press the "X" button, you will close Webztral.

- If you press and hold the Webztral window topbar, you can move its window.

//The same logic for resizing, opening, closing, minimizing, maximizing and moving windows is used in the windows of "About", "Webztral", "Docstral", "File manager" and "Astral projection".

1.2. Webztral top toolbar and bookmarks bar:

- If you press the "←" button, you will be taken to the previous page of the history.
- If you press the "→" button, you will be taken to the next page of the story.
- If you press the "Home" button, you will be taken to google.com.
- If you click on the "Address bar", you can enter a link and open it by pressing "Enter" or the "Go →" button.
- If you click on the "Add a bookmark" button, a window will appear with writing fields and the "Save" and "Cancel" buttons, from where you can add a bookmark or stop adding.
- If you right-click on a bookmark, a window will appear with writing fields and the "Delete", "Cancel" and "Save" buttons, where you can uncheck or edit a bookmark.
- If you click on a bookmark, it will open it inside Webztral.
- If you click the "Outside" button, you will open the current page in a new tab on the real browser.
- If you click on the "History" button, a drop-down menu will appear with the most recently visited sites. From here you can see the browsing history for the current session.
- If you click "Delete", you can delete pages from the history.
- If you press "Clear all", you can clear the entire history for that session.
- If you press the "Export" button, a drop-down menu with export buttons will appear.
- If you press the "Selected to png" button, you can select an area from the visible screen by marking, then the "Save" and "X" buttons appear and you can save it as a png or decline.
- If you press the "Visible to png" button, you can download the visible page as a png.
- If you click the "Full to png" button, you can download the entire page as a png.

2. Docstral

-Docstral window bar:

//It uses the same logic for resizing, opening, closing, minimizing, maximizing and moving as on the other windows, but there are additional buttons in this bar.

- If you press the "Zen Mode" button, you will hide the top toolbar "1", the top bar "2", the bottom bar "2" and the bottom bar "1", allowing you to focus better at work by removing unnecessary eye strain.
- If you press the "Planchette Mode" button, you will open the main part of Docstral, where you can work with text files.
- If you press the "PDF mode" button, you will open a Docstral extension where you can manage .pdf, .jpg .png files.
- If you hit the "QR generator" button, you will open a Docstral extension where you can create QR codes.

2.1. Docstral – Planchette Mode

// "Plachette mode" has two upper and two bottom bars and a window bar.

// "Pdf mode" and "QR generator" only have a window bar.

2.1.1. Docstral top toolbar „1“

- If you press the "<" button, you will open the previous document from the list of documents in the "document bar".
- If you press the ">" button, you will open the next document from the list of documents in the "document bar".
- If you press the "Import" button, you can import text files in .docx, .txt, .jdoc.json format.
- If you press the "+" button, you can create a new .jdoc.json text file (document).
- If you press the "Document bar" button, a drop-down menu will appear with a list of all available documents from the planchette_mode folder in the file manager, with a rename button and a document delete button to the right of each document.
- If you click on a document from the list, you will open it.
- If you press the "Rename" button, you can rename the document.
- If you press the "Delete" button, you can delete the document.
- If you press the "-" button, you can delete the document that is open.
- If you press the "Export" button, a drop-down menu will appear with 5 export buttons in .docx, .pdf, .txt, .jdoc.json, .png format.
- If you press the "U" button, you will return the previous state of the document (Ctrl + Z).
- If you press the "V" button, you will return the current state of the document (Ctrl + Y).

2.1.2. Docstral top toolbar „2“

- If you press the "-" button, you will zoom out the document.
- In the zoom bar field, you can see the zoom. If you enter a number (from 0 to 250%) in its writing field and press "Enter", you can change the scale of the document.
- If you press the "+" button, you can upscale the document.
- If you click the "Find and Replace" button, the search and replace sidebar in the document will open on the left.
- If you press the "Clipboard" button, the clipboard sidebar will open on the left.
- If you press the "Bold" button, you can bold the font to marked text.
- If you press the "Italic" button, you can tilt the font to marked text or write in italics on the line.
- If you press the "Underline" button, you can underline marked text or write underlined line text.
- If you press the "Strikethrough" button, you can strike out marked text or write strikethrough text on the line.
- If you press the "Superscript" button, you can make marked text as superscript or write with superscript on the line.
- If you press the "Subscript" button, you can make marked text as a subscript or write with a subscript on the line.
- If you press the "Clear Formatting" button, you can remove the formatting of marked text, and this will remove all: bold, italic, underline, strikethrough, superscript, subscript, font, font colour, size, Heading 1, 2, 3, 4, Title, Quote, Align left, center, right and justify. The marked text will become TEXT layer, Align left, in black, Times New Roman, 12pt.
- If you press the "Astral projection" button, you can add marked text from the document to the astral projection window.

- If you press the "Eagle View" button, the eagle view sidebar will open on the right.
- If you press the "Tables" button, the sidebar for the tables will open on the left.
- If you press the "Font" button, a drop-down menu with fonts will appear, from where you can change the font of marked text.
- If you press or hold the "v" button, you can lower the font size of marked text or line.
- If you press or hold the "^" button, you can increase the font size of marked text or line.
- If you press the "Text colour" button, a window with standard colours and an option for custom RGB colour will appear, from where the colour of marked text or line can be changed.
- In the "Current font and size" cell, you can see the font and size of marked text. If the marked text is uniform, it will be displayed, for example: "Times New Roman, 12pt", but if there are parts in the marked text that are of different size or font, the cell will display: "Mixture of Fonts".

2.1.3. Docstral bottom toolbar „2“

- If you press the "Layers" button, the sidebar for the layers will open on the left.
- If you press the "H1" button, you can make marked text as a Heading 1 layer.
- If you press the "H2" button, you can make marked text as a Heading 2 layer.
- If you press the "H3" button, you can make marked text as a Heading 3 layer.
- If you press the "H4" button, you can make marked text as a Heading 4 layer.
- If you press the "TTL" button, you can make marked text as a Title layer.
- If you press the "Q" button, you can make marked text as a Quote layer.
- If you press the "TXT" button, you can make marked text as a Text (base) layer.
- If you click the "Align Left" button, you can align text to the left of the page.
- If you press the "Center" button, you can center text in the middle of the page.
- If you press the "Align Right" button, you can align text to the right of the page.
- If you press the "Justify" button, you can align text to the left and right of the page at the same time.
- If you press the "Lines" button, the row sidebar will open on the left.
- In the "Current style" cell, you can see the style (layer) of marked text.

2.1.4. Docstral bottom toolbar „1“

- If you press the "<<" button, you will be taken to the first line in the current document.
- If you press the "<" button, you will go to the previous line.
- If you press the ">" button, you will go to the next line.
- If you press the ">>" button, you will go to the last row.
- If you type a number in the write box on the Docstral navigation bar and press "Enter", you will be taken to the line with the number entered. You can see the total number of rows in the current document and the order you are currently on.
- If you enter a prompt in the integrated writing field and press "Enter", the language model will generate a response in the document where the caret is currently located.
- If you press the "Content" button, the content sidebar will open on the left.
- If you press the "Info" button, the information sidebar will open on the left.
- If you press the "Print" button, you can print the document.

-Docstral sidebars:

2.1.5. Docstral Placnhette mode sidebar (Home screen)

- If you press the "X" button, you can close the sidebar.
- If you click the "New document" button, you can create a new .jdoc.json text file (document).
- If you click on one of the documents in the "recent files" list, you can open it.

2.1.6. Find and Replace Sidebar

- If you click on the "Find what" text box, you can enter search text.
- If you click the "Replace with" text box, you can enter text to replace.
- If you press the "Find" button, you will search for the entered text from the "Find what" field.
- If you click the "Replace all" button, you will replace the text of all found search results with the text entered from the "Replace with" field.
- If you click the "Replace Selected" button, you will replace the text of only the selected search results that are marked with a checkbox in the list of results with the text entered from the "Replace with" field.
- If you press the "←" button, you will go to the previous result from the list of finds.
- If you press the "→" button, you will be taken to the next result from the list of finds.
- If you click on the main "Select all" checkbox on the list of found results, you can mark or unmark all results.

To the right of the "Select all" checkbox, you can see the total number of results found and the number of results marked with a checkbox.

- If you close the sidebar, the background coloring of the text (found results) in the document will disappear.

2.1.7. Clipboard Sidebar

- If you press the "Clear all" button, you will clear all cells with copied text in the clipboard.
- If you press the "Copy" button, you will copy the text from the cell.
- If you press the "Regen" button, a writing field will appear, where you have two options: In the first option, you can enter a prompt and press "Enter", after which a response will be generated in the document. In the second option, you can press the "Regen" button again to hide the write field.
- If you click on the "Delete" button, you will delete the text cell.

2.1.8. Eagle View Sidebar

- If you place the mouse on the sidebar of "Eagle view", a small window will appear on the right, in which about 50 lines of the document are visible on a small scale.
- If you move the mouse vertically (up and down) in the sidebar, you can observe the entire document from above.
- If you stop moving the mouse and press the sidebar, you will freeze the view of the preview window.
- If you place the mouse on part of the preview window, a rectangular window (magnifying glass) will appear, in which about 7 lines of the document are visible on a larger scale in real time, thus the text is easy to read.
- If you press the window (magnifying glass), you will freeze its view.

//When freezing the view in the window and in the magnifying glass, you can scroll and work elsewhere in the document while observing the text in the magnifying glass.

2.1.9. Tables Sidebar

- If you enter numbers in the "Rows" and "Columns" writing fields, then press the "Insert table" button, you will create a new table with as many rows and columns on the row that was last clicked.
- If you press the "↑" button, you will add a new row above a marked cell in the table.
- If you press the "↓" button, you will add a new row under a marked cell in the table.
- If you press the "←" button, you will add a new column to the left of a marked cell in the table.
- If you press the "→" button, you will add a new column to the right of a marked cell in the table.
- If you click the "Delete row" button, you will delete the row of the marked cell from the table.
- If you click the "Delete column" button, you will delete the column of a marked cell from the table.
- If you click the "Delete table" button, you will delete the table in which the cell is marked.
- In the "Marked cells for context" field, you can see the cells that are currently marked and from which the language model can use as context for a response (e.g. A1, B2).
- In the "Enter a prompt here" field, you can enter a prompt and if you press "Enter" or the "Generate answer" button, a response will be generated with the context of the marked cells, on the row below the table.
- If you click on the "Explain data" button, a response from the language model will be generated, on the row below the table, which contains an explanation of the entire table.
- If you click "Export to python pandas", a python code will be generated on the row below the table. Through the generated python script, a table in .xlsx format can be created when running through a terminal.
- If you press the "Ruler" button, you can measure lengths in millimeters in the document.

2.1.10. Layers Sidebar

- If you click the "Select all layers" checkbox, you can select or unmark all layers from the list.
- If you click the "Toggle visibility for all" button, you can hide the contents of all layers of the document.

//Hiding layers doesn't erase content, but makes it invisible by rearranging the lines of the entire document

- There are 15 layers in the sidebar: Heading 1 All, Heading 1 Names, Heading 1 Content, Heading 2 All, Heading 2 Names, Heading 2 Content, Heading 3 All, Heading 3 Names, Heading 3 Content, Heading 4 All, Heading 4 Names, Heading 4 Content, Title, Quotes and Content (text and tables). On the right of each layer there is a button "Ask Magi for this layer" and "Toggle visibility", and on the left there is a checkbox.

//All – Title Names, Text and Tables, Names – Title Names Only, and Content – Text and Tables for the selected "Heading" only.

- If you press the "Ask Magi for this layer" button, a writing field will appear in which you can enter a prompt and if you press "Enter", a response will be generated to the line that was pressed last, taking only the text from the selected layer as context.
- If you press the "Toggle visibility" button, you can hide the text of this layer of the document or restore it back.

- If you click on the "Layer actions" button, a drop-down menu with buttons (options) will appear with three tabs: Section "Create outside – New .jdoc", which contains the buttons "From selected layers" and "From all layers", Section "Create inside – New version", which contains the buttons "From selected layers" and "From all layers", Section "Other actions", which contains the buttons "Export selected to .txt file" and "Remove selected from here".
 - If you click the "From selected layers" button (in the create outside section), you will create a new .jdoc.json document that contains only the text of the selected layers.
 - If you click the "From all layers" button (in the create outside section), you will create a new .jdoc.json document that contains the text of all layers.
 - If you click the "From selected layers" button (in the create inside section), you will create a new version of the document in the same .jdoc.json document you are working on, and the new version will contain the text of the selected layers only.
 - If you click the "From all layers" button (in the create inside section), you will create a new version of the document in the same .jdoc.json document you are working on, and the new version will contain the text of all layers.
- //When creating a new document or a new version, a window appears in which you need to enter their name.
- //After you create a new version of the document, it appears in the "Document versions" list in the sidebar below the layer list. You can switch document versions by clicking on the version cell in the list. Thus, a document may have a separate version for a different language, or a document may have documents associated with it in the form of different versions. That is, a new version can be created from one layer, but the text does not have to be the same as in the main version.
- If you click the "Export selected to .txt" button, you can download the text of the selected layers to a text file.
 - If you click the "Remove selected from here" button, you will delete the text of the selected layers from the document and the lines will be rearranged.
- //In Docstral, each line is a paragraph and there are no indentations.
- //Versions in Docstral resemble "Git" for writers.

2.1.11. Lines Sidebar

- If you enter a number (number of lines) in the writing field, then from the checkboxes you mark the line numbers and click the "Add before" button, you will add the specified number of new lines before the selected ones.
- If you enter a number (number of rows) in the writing field, then select the line numbers from the checkboxes and click the "Add after" button, you will add the specified number of new lines after the selected ones.
- If you click on the "Delete all" button, you will delete all lines and their contents.
- If you click on the "Delete selected" button and select rows from the list through the checkboxes, you will delete the selected rows and their contents.
- If you enter the line numbers (for example, 1, 2, 5-10) in the "Remove specific" field and click the "Delete" button, you will delete these lines and their contents.
- If you press the "All" checkbox, you can select or unmark all the lines from the list in the sidebar.
- If you press the "Even" checkbox, you can select or unmark all even rows from the list in the sidebar.

- If you press the "Odd" checkbox, you can select or unmark all odd lines from the list in the sidebar.
- Each line in the list has a "Delete" button. If you press the "Delete" button, you will delete its order and contents.
- Each line in the list is like a cell that contains a checkbox, a sequence number of the row, part of the text of each individual line, the layer of the line text and the delete button.

2.1.12. Content Sidebar

- If you enter a line number from the document in the "Add before line" writing field, then select the content items from the list through the checkboxes and click on the "Create Table of Contents" button, the content of the document will appear before the specified line in the writing field.
- The content creation list includes 5 things: Titles, Headings 1, 2, 3, and 4.

2.1.13. Info Sidebar

- The information sidebar contains two data cells that are constantly changing during operation.
- The "Structure counts" list shows number data: Title blocks, Headings 1, Headings 2, Headings 3, Headings 4, and Quotes.
- Списъкът "Word Distribution" показва данни за брой: Total words, Words in Titles, Words in Headings 1, Words in Headings 2, Words in Headings 3, Words in Headings 4 и Words in Quotes.

2.1.14. Docstral workspace:

Buttons when marking text in Docstral:

- If you mark text in Docstral, a drop-down menu with buttons will appear below it.
- If you press the "Bold" button in the button menu, you can bold the font to marked text.
- If you press the "Italic" button in the button menu, you can tilt the font to marked text or write in italics on the line.
- If you press the "Underline" button in the button menu, you can underline marked text or write underlined line text.
- If you press the "Strikethrough" button in the button menu, you can strike out marked text or write strikethrough text on the line.
- If you press the "Superscript" button in the button menu, you can make marked text as superscript or write in line superscript.
- If you press the "Subscript" button in the button menu, you can make marked text as a subscript or write with a subscript on the line.
- If you press the "Clear Formatting" button in the button menu, you can remove the formatting of marked text.
- If you press the "Astral projection" button in the button menu, the marked text will be placed in a new cell in the "Astral projection" window.
- If you press the "Cut" button in the menu, the marked text will be cut and pasted into the Docstral clipboard.
- If you press the "Copy" button in the menu, the marked text will be copied and pasted into the Docstral clipboard.
- If you press the "Paste" button in the menu, copied text will be placed in place of the marked text.
- If you press the "Find Here" button in the menu, the "Find and Replace" sidebar will open and the marked text will be placed in the "Find what" writing box, then search results will be displayed in the sidebar list.

- If you press the "Find Inside" button in the menu, the internal browser "Webztral" will open and the marked text will be searched in the google.com
- If you press the "Find Outside" button in the menu, a new tab of the real browser will open and the marked text will be searched in it.
- If you click on the typing box in the menu, then enter a prompt and press "Enter", text will be generated in place of the marked text.
- If you move the mouse away from the button drop-down menu, it will fade and eventually disappear. If you click on the side of the screen, the button drop-down menu will disappear.

2.2. Docstral – PDF Mode

- There is a sidebar on the left. On the right is the work field.
- If you click on the "Drag and drop files" field in the sidebar or move a file from your computer, you can upload files in "Pdf mode". Uploaded files will appear in the list of files in the sidebar on the left, and their pages will appear in the workspace in the background. Several .pdf, .jpg, .png files can be uploaded, and in the workspace all .pdf files are displayed as separate pages that are numbered and on which the name of the original .pdf document is written.
- If you press the "X" button to the right of a document from the list, you will delete it from the list and remove its pages from the work field.
- In the "Selected" field in the sidebar, you can see the number of marked pages in the work field.
- From the checkbox in the sidebar, you can choose whether .pdf the document to be exported at the end has the page number (Pagination).
- In the writing box in the sidebar, you can enter text to appear as a watermark on each page in the exported document.
- If you click the "Merge all to pdf" button, you will merge all pages from the workfield into one .pdf document and it will be downloaded.
- If you click the "Merge selected to pdf" button, you will merge only the marked pages from the workspace into one .pdf document and it will be downloaded.
- If you click the "Split all to pdf" button, you will download all pages from the workspace as separate .pdf files that are collected in one .zip file.
- If you click the "Split selected to pdf" button, you will download the marked pages from the workfield as separate .pdf files that are collected in one .zip file.
- If you click the "Split all to png" button, you will download all pages from the workfield as separate .png files that are collected in one .zip file.
- If you click the "Split selected to png" button, you will download the marked pages from the workspace as separate .png files that are collected in one .zip file.
- If you press and hold on the workbox, you can mark pages via "Drag selection".
- If you click once on an unmarked page, you'll mark it.
- If you click once on a marked page, you'll unflag it.
- If you click on a page and hold, you can move the location of this page to another location in the document, and when you export it, it will be on that serial number.

- If you double-click on a page, you will open .pdf viewer, where you can scroll through all the pages in the workspace as one document.
 - If you press the "Zoom in" button in the .pdf viewer, you will zoom in on the document.
 - If you press the "Zoom out" button in the .pdf viewer, you will downsize the document.
 - If you press the "X" button in .pdf the viewer, you will exit the viewer.
 - If you place your mouse on a page in the workbox, the "Rotate left", "Rotate right" and "Delete" buttons will appear above the page.
 - If you press the "Rotate left" button, you will rotate the page to the left (-90°).
 - If you press the "Rotate right" button, you will rotate the page to the right (90°).
- You can rotate each individual page 360° and it will be visualized in this way in the workspace and in the exported document or photo.
- If you click the "Delete" button, you can delete a page from the work field.
 - If you mark a page (or pages), the "Rotate left -90°", "Rotate right 90°", "Delete selected" and "Cancel" buttons will appear
 - If you press the "Rotate left -90°" button, you will rotate all marked pages to the left.
 - If you press the "Rotate right 90°" button, you will rotate all marked pages to the right.
 - If you click the "Delete selected" button, you will delete all marked pages from the work field.
 - If you press the "Cancel" button, you will unmark all the pages and the buttons will disappear.

2.3. Docstral - QR generator

- There is a sidebar on the left. On the right is the work field.
- The workspace is empty. The visualization of the QR code in the workspace appears only after generation.
- If you click on the writing field, you can write text or paste a link into it.
- If you press the "Generate" button, a QR code will be generated in the workbox.
- If you press the "Reset" button, you will remove the generated QR code from the workbox.
- If you change the position of the slider, you can choose the pixel size for the QR code.
- If you press the "Dot style" cell, a drop-down menu will appear with five buttons from which you can select one of these styles for the QR code. The possible options are: "Square style", "Dot style", "Rounded style", "Strongly rounded" and "Elegant style".
- If you click on the "Code colour cell", you can choose the color of the code points.
- If you click on the "Background" cell, you can choose a background color.
- If you press the "Select logo" button, you can select a logo from your computer, which will be in the center of the QR code (.png or transparent png).
- If you click on the "Export options" button, a drop-down menu with 4 buttons will appear.
- If you press the "Export png" button, you will download the QR code as a png.
- If you click the "Export transparent png" button, you will download the QR code as a transparent png.
- If you click the "Export svg vector" button, you will download the QR code as an svg vector.
- If you press the "Export jpg" button, you will download the QR code as a .jpg.

3. IMVISTRAL

- Under development, at the moment its design is a combination of media player and chat.

4. FILE MANAGER

4.1. File manager window bar

- The same as the other windows.

4.2. File manager top toolbar „1“:

- The main directory is the root system folder.
 - If you press the "←" button, you will be taken to the previous folder of the file manager.
 - If you press the "→" button, you will be taken to the next folder of the file manager.
 - If you press the "↻" button, you will refresh the directory.
 - If you press the "Home" button, you will return to the system folder if you are in another folder, or refresh it if you are in it.
 - If you click on the "Search bar", you can find files by name.
 - If you click on "New System Layer", you can create a new memory layer.
 - If you press the "Import" button, you can add files from your computer to the file manager.
 - If you press the "Export" button, a drop-down menu with export buttons will appear.
 - If you press the "Export all" button, you will download all files from the current folder to a zip file.
 - If you click the "Export selected" button, you will only download the selected files with a checkbox from the current folder.
 - If you press the "Delete" button, a drop-down menu with delete buttons will appear.
 - If you press the "Delete all" button, you will delete all files from the current folder.
 - If you click the "Delete selected" button, you will delete only the selected files with a checkbox from the current folder.
 - If you click the "Settings" button, you can add language models to the ollama list and use them in the engine model (if they are pre-downloaded).
 - If you click on the main checkbox, you can mark or unmark all files in the current folder.
 - If you click on a checkbox in a file, you will only mark or unmark that file.
 - If you press the "Delete" button, you will only delete this file.
 - If you click on a file .txt, .json, .jsonl, .py, .html, .js, .css, or any other format, it will open on the right side of the file manager.
 - If you open a text file on the right side of the file manager, you will be able to edit it, with the "Cancel" and "Save" buttons appearing at the bottom of the file manager.
 - If you mark text in the file manager, the "Copy", "Ask here", "Regen Selected" and "Expand selected" buttons will appear.
 - If you press the "Copy" button, you can copy the text, but it will not appear in the Docstral clipboard.
 - If you press the "Regen Selected" or "Expand selected" button, a window will appear with a prompt input box and the "X", "Cancel" and "Execute" buttons.
- //On the left side of the file manager, the name, file format, and size of each file are displayed.
- If you press and hold the border between the left and right sides of the file manager, you can expand and shrink the left and right sides.

5. ASTRAL PROJECTION

5.1. Astral projection window bar:

- If you click on the "Astral projection" button on the main page, it will open its window.
- If you press the "minimize" button, the window will hide and the icon will reappear on the main page.

//In astral projection, there is no "X" button to turn off, and it is constantly active. Its icon in a minimized state is not a bubble like the apps, but a square.

//In this bar, you can see information about the number of cells and the words that are available in it.

- Resizing and moving the astral projection window is done in the same way as on the other windows.

5.2. Astral projection top toolbar

- If you press the "+ New cell" button, you will add a new cell under the last cell.
- If you press the "Manifest selected" button, the text from the selected checkbox cells will be placed in the "chat container" on the main page, where you can write an additional prompt before generating a response.
- If you press "Manifest all", the text from all cells will be placed in the "chat container" on the main page, where you can write an additional prompt before generating a response.
- If you press the "Clear all" button, you will remove all cells and their contents.
- If you press the "Mark/unmark all" checkbox button, you can mark or unmark all cells.

5.3. Astral projection workspace:

- If you place the mouse above or below a cell, a "Add new cell" button will appear.
- If you click the "Add new cell" button, you can create empty cells (between existing ones).
- If you press and hold a cell, you can drag it around.
- If you click on the cell name, you can rename it.
- If you click on the write field of the cell, you can edit and add text manually.
- If you press the "Copy" button, you will copy the contents of the cell, but it will not appear in the Docstral clipboard.
- If you press the "Manifest" button, you will paste the text of this cell into the "Chat container" on the main page and you will be able to write an additional prompt before generating a response.
- If you click the "Delete" button, you will delete the current cell.
- If you click on the checkbox of the cell, you can mark or unmark this cell.

III. Magixtral - System layers, workmodes and logic

- "Magixtral", "Midextral" and "Olachra" are three separate software systems that form a common ecosystem. In this case, it is not about the "Magi" operating system, but about the three separate platforms considered as systems.
- In every system, there is an interaction between a user and a system.
- In each ecosystem, there is an interaction between user and ecosystem.
- With each interaction, receiving and transmitting processes take place.
- The principles of interaction are the same at the micro and macro level.

Receiving and transmitting processes in the interaction between the user and the magixtral system, depending on the point of view:

From a user's point of view:

- Work: Layers, working modes and logic (System Layers);
- Understanding: Applications of "Magixtral" (Appstrale);
- Learning: Main page of "Magixtral";

From a user's point of view:

- The user works (transmits) to the layers and operating modes (System Layers).
- The user understands (receives and transmits) through the applications of "Magixtral" (Appstrale).
- The user learns (receives) from the main page of "Magixtral".

From the point of view of the system:

- Learning: Layers, working modes and logic (System Layers);
- Understanding: Applications of "Magixtral" (Appstrale);
- Work: Main page of "Magixtral";

From the point of view of the system:

- The system learns (receives) from the layers and operating modes (System Layers).
- The system understands (receives and transmits) through the applications of "Magixtral" (Appstrale).
- The system works (transmits) to the main page of "Magixtral".

//According to the principle of Hermeticism: What is below (in the system), is also above (in the ecosystem).

Receiving and transmitting processes in interaction between user and ecosystem, depending on the point of view:

From a user's point of view:

- Work: "Olachra";
- Understanding: "Midextral";
- Learning: "Magixtral";

From a user's point of view:

- The user works (transmits) to "Olachra".
- The user understands (receives and transmits) through "Midextral".
- The user learns (receives) from "Magixtral".

From the point of view of the ecosystem:

- Learning: "Olachra";
- Understanding: "Midextral";
- Work: "Magixtral";

From the point of view of the ecosystem:

- The ecosystem learns (receives) from "Olachra".
- The ecosystem understands (receives and transmits) through "Midextral".
- The ecosystem works (transmits) to "Magixtral".

Additional information about the ecosystem:

1. "Magixtral"

- It is the platform that serves as the operating environment for the "Magi" operating system, its "Appstrale" applications and the language model.
- The name of this platform consists of the name of the operating system "Magi", which symbolically means "magus/wise mens" and the word "astral", which means "starry/stellar". That is, the name of the platform means "starry magus/stellar wise men".

1.1. "Mistral artificial general intelligence (Magi)"

- It is a web operating system that works with the applications in the "Appstrale" and the language model.

- The name of this operating system is an abbreviation of "Mistral Artificial General Intelligence - MAGI", which has a literal meaning.

//The name is close in meaning to that in the Japanese anime "Neon Genesis Evangelion", in which "MAGI" is a system of three supercomputers, which is a random coincidence in names with the "Magi" operating system. Another coincidental coincidence is the word "Magga", which in Pale language means path and symbolizes the path to enlightenment.

//The name "Neon Genesis Evangelion" means "Good News for the New Beginning" from the Greek "Neon", meaning "new", "Genesis" means "generation/beginning", and "Evangelion" means "good news".

Appstrale (applications):

- The name of the applications in the operating system "Magi" contains the words "App" and "stral". The word "App" means "application" in English. The word "stral" means "rays" from the Dutch language (stralen). That is, the name "Appstrale" literally means: "Ray applications". Symbolically, it means that every application is a ray of light/enlightenment.

- In the "Magi" operating system, there are four applications and one addon, which are its "five elements". The applications are called "Webztral", "Docstral", "Imvistrat" and "File manager", and the extension is called "Astral Projection".

1.1.1. Webztral – Internal browser of Magi OS

- The name of this application literally means: "web (browser) ray".

- Working Principle:

Webztral does not load web pages directly into the local (external/real) browser (e.g. via `<iframe>`), but works on the principle of video streaming on a virtual machine that resides on the server.

Webztral Backend:

When you run Webztral, the backend (Python/FastAPI) runs a hidden instance of a Chromium browser via the Playwright library. This browser works in an isolated server environment with a fixed resolution (1280x720 pixels HD). It is loaded with special flags (such as `--disable-blink-features=AutomationControlled`) to bypass bot protections and behave like a normal user browser. The connection between the interface and the server is made through WebSockets, which allows two-way communication without delay.

The server uses the Chrome DevTools Protocol (CDP) to take continuous screenshots (screencast) of what is happening in the virtual browser. Photos are compressed as JPEG frames (with a certain quality and frame skip - `everyNthFrame: 3` to save resources) and sent to the frontend.

-The browser saves a separate history for each session in the .json file of that session. When the browser is used without creating a session in active mode or in passive mode, history is not saved.

Webztral Frontend:

In the user interface, Webztral uses an HTML5 `<canvas>` element rather than standard web components. When the JavaScript frontend receives a new JPEG frame through WebSocket, it simply "draws" it on Canvas. This looks like a real browser, but it's actually an interactive video.

When you move the mouse, click, scroll, or type on the keyboard on Canvas, JavaScript functions capture these actions. They calculate the exact coordinates of the mouse relative to the scale of the Canvas and send them back to the server as text commands (e.g. `{"action": "click", "x": 500, "y": 300}`). The server's playwright receives these commands and "simulates" the same click or keystroke in the virtual Chromium. The result (page change) is captured and sent back as a new frame.

Webztral is special in the "Magi" operating system, because of its connection with artificial intelligence:

- Reading Context: The application always extracts all the text from the loaded web page (`get_full_context`) and feeds it as the context of the AI model (Ollama/LLM). For this reason, when there is an open page in Webztral and a prompt is entered in the chat, the language model responds as if it sees the page that is open.

- Interactive menus: When you mark text in the video stream (Webztral), the server executes `window.getSelection()` in the virtual browser and returns the marked text to the local interface. This allows you to press buttons such as "Ask Here", "Explain" or "Summary", which directly feed the AI with the information marked by the website.

There are also stability mechanisms in the code:

- Drop Frames (Backpressure): If the connection slows down, the server stops sending new frames so as not to overload memory (Memory Leak).

- Defibrillator: If the virtual browser "crashes", the system automatically creates a new context (context.new_page()) and tries to "revive" the view unnoticed by the user.
- Heavy caching: On each startup, the server deletes accumulated GPU and Code cache from Chromium to keep the system fast, otherwise large cache will accumulate.

To put it simply: Webztral is a remote browser, streamed as video to HTML5 Canvas, whose commands are transferred in real time, allowing for full and safe integration of external web content with the local AI assistant. At the heart is the principle of Remote Browser Isolation.

1.1.2. Docstral

- The name of this application literally means "document ray" doc + stral.

Unlike standard text editors that simply keep raw HTML or plain text, Docstral is a Block-based architecture with a strong emphasis on semantic analysis, layer management, and AI integration.

Data Block Model (JSON-based format):

Docstral does not save documents as traditional files, but as structured JSON objects with an extension .jdoc.json format. Instead of seeing the entire text as a whole, the system breaks it down into separate "blocks". Each paragraph, title, citation, etc., is a separate entity with its own unique ID, type, and content. This allows the system (and artificial intelligence) to precisely manipulate specific parts of the document without affecting the rest.

Real-time synchronization via (DocstralSync):

- The application uses a module called DocstralSync. It works as a "translator" between what is seen on the screen (HTML DOM) and what is stored on the server (JSON):

-Listen for changes: When you type in the editor (contenteditable="true"), the system captures the changes (via MutationObserver and input events).

-Extraction: Extracts the HTML elements, cleans them up, and converts them into JSON blocks.

-Debounce write (Auto-save): Waits for a short moment (e.g. 1.2 seconds) after you stop typing, and automatically sends blocks to the Python backend (/api/docstral/render and /api/docstral/save) to save them without the need to manually press the "Save" button.

Semantic Layers (DocstralLayerManager):

-Docstral treats text the same way a graphic software like Adobe Photoshop treats images – that is, through layers.

When you type, the system automatically categorizes the text depending on how you mark it: Headings go in one layer, quotes in another, and plain content in thirds. There are a total of 15 layers.

-You can "hide" or "show" these layers by injecting CSS rules. Each time the visibility of a layer changes, the visibility status of the .jdoc.json file also changes, i.e. the layers use code in the backend and frontend.

-AI Application: You can select a specific layer (for example, "Heading 1 Content" only) and have the AI read or rewrite *only* that logical part of the document. You can also generate an entirely new document containing only selected layers.

Inline Editing: If you mark text, you can call the AI to expand or rewrite it. The server returns the response as a stream and it is written letter by letter directly into the document (via the `typeWriter` function).

AI TableManager (TableManager): There is a dedicated assistant for tables. If you tag cells and you can ask the AI to "explain the data" or generate Python/Pandas code to export that particular table.

Local Undo/Redo: DocstralHistory keeps up to 50 steps back in time, recording not just the text, but also the cursor position (caret position) to bring you back exactly to where you were.

Clipboard Time Machine: Keeps a history of everything (except in some places) that has been copied or cropped, allowing you to restore old pieces of data.

In the Docstral application, the clipboard is mainly designed for input (data reception), unlike the "Astral projection" extension, which is mainly designed for output (transmitting data to the chat), but both can receive and transmit information.

Zen Mode: How it works: When you activate the `toggleZenMode` feature, JavaScript dynamically calculates the height of the top and bottom toolbars and shrinks them to 0 pixels. Through smooth CSS transitions, the interface literally "disappears", leaving only the document on the screen.

This working mode is only active in "Planchette mode" and is based on "zen" from Buddhism. The name "zen" comes from the word "dhyana", which means "contemplation" in Sanskrit. Contemplation is a state of continuous, calm and concentrated observation of an object (in the case of a document), accompanied by deep reflection and admiration. This mode is useful when you need to concentrate on documentary work, without distractions with background elements.

Planchette Mode: How it works: This is the main mode of working with a file `.jdoc.json` format, it uses DocstralSync and the block architecture, which allow the user and the AI to write in symbiosis. The term hints at the philosophy of the application, because in it you don't just "enter data", but "channel" ideas through artificial intelligence.

The name "Planchette" (from French: *planchette* - "board") means a small object in the shape of a heart or tear, which is usually made of wood or plastic, supported by wheels (or feet) and often with a pencil hole. It is known in spiritualism. The planchette is known as an automatic writing tool associated with spiritualistic séances. It is believed that through a light touch by people (mediums), The board moves on its own and writes messages from "ghosts". In the case of Docstral, the user can be thought of as the "medium" and the language model as the "spirit."

PDF Mode: Uses the pdf.js library to turn PDFs into interactive images of the workflow. Allows splitting, joining, turning pages, and watermarking.

QR Mode: Built-in QR code generator with styling ability (add logo, change points, export to SVG/PNG).

The ".jdoc.json" file format is characteristic of the Docstral application "Planchette Mode" and indicates that it is a specific document created in the "Magi" operating system.

//**.json** finally, ensures that any operating system (Windows, Linux, macOS) and any text editor (VS Code, Notepad++) will recognize the file as valid structured text and open it without errors.

//**.jdoc.json** turns a document from a simple document into a dynamic database.

//This is an example of a standard .jdoc.json file:

```
{
  "file_version": "3.0",
  "metadata": {
    "id": "doc-1777843918",
    "name": "New Document 1",
    "active_version": "main",
    "created_at": ".....",
    "last_modified": "....."
  },
  "visible_layers": [
    "h3_names",
    "h3_content",
    "h4_names",
    "h4_content",
    "title",
    "quotes",
    "h1_names",
    "h1_content",
    "h2_names",
    "h2_content",
    "content"
  ],
  "blocks": [
    {
      "id": "b-1778017779652366",
      "type": "paragraph",
      "content": ".....",
      "layer": "base"
    }
  ],
  "versions": {
    "v_4009": {
      "created_at": ".....",
      "last_modified": ".....",
      "visible_layers": [
```



```

        "h1_names",
        "h1_content",
        "content"
    ],
    "blocks": [
        {
            "id": "v1778360604009-0-116",
            "type": "h1",
            "content": ".....",
            "layer": "base",
            "style": ""
        }
    ]
}
}
}

```

//In the above example, this file contains two documents (two versions), both versions have one line (paragraph ".....") and there is a difference in the selected visible layers and the marking of the row (paragraph) as a layer. For each new row, a new block is created in the file.

1.1.3. Imvistrat

- This application serves to generate photos, video and sound. The name "Imvistrat" is formed from image, video, sound (im + vi + s) + tral (ray), which means "media ray".

//Currently, two main methods are used to generate media (photos, video and sound): generation by diffusion and generation by autoregression. Simply, it can be said that their similarity is that both methods are probabilistic models, which means that they do not know the "exact" answer, but choose the most likely next state according to what they have learned. The difference is that through the autoregressive method, new content is "built" step by step, with each new step adding information to what has already been created based on it. Whereas through the diffusion method it starts from full noise and "eliminates the uncertainty" step by step until a clear structure remains.

//By way of logic, this is reminiscent of the "Uncertainty Principle" in the quantum mechanics, but presented in a virtual environment. This similarly means that for the same input situation (prompt) there can be many possible correct outputs (responses), which is why it can be said that these methods of media generation are based on inductive uncertainty/probabilities. The reason for this is that in quantum physics, before a particle is measured, it is in a "superposition" (multiple possible states). With probabilistic AI, it's exactly the same, because before the model generates the final pixel (photo/video) or sound, the response to the prompt exists as a wave of probabilities.

//A few years ago, during a research (and learning) about the principle of operation and use of an oscilloscope, I noticed that it is not limited to simply visualizing electrical signals. Then I came across a video that presented a visualization of the principle of operation, which is similar to the one currently used in Imvistrat.

//Example: In this video [<https://www.youtube.com/watch?v=4gibcRfp4zA>], in the period (1:25 – 2:45), the visualization is presented in the video, but the generation of media through Imvistral is done differently from the way in this video, and in addition to simple shapes (vectors) it is possible actual images and videos (rasters) to be generated.

- With "Imvistral" a method of frequency media generation (Frequency Media Generation - FMG) is used. Through this method, image and video from sound can be generated, or sound from image and video can be generated.

//At the beginning of the project, the "Webztral" application was not planned for development, but was added due to the need for a working code for streaming environment. Due to the complexity of "Imvistral", my decision was to separate the tasks and before developing "Imvistral", to make something simpler that would work in a similar way (simply put, looking at vectors/data with the language model + computing and streaming images, video and sound from the application to a screen). For now, the streaming code from "Webztral" is the basis for transmitting the generated results of "Imvistral" to its window.

- The principle of frequency media generation is based on the transformation processes of logical induction and deduction, in which:

1. With inductive frequency media generation, it is possible to perform sound into light conversion.

//With this media generation, two options are currently being developed: in one option, the language model can generate images or videos after a prompt is entered in the "Chat container" of "Imvistral". Each prompt produces a mixture of "sound waves" of different frequencies in vector form, which the language model uses from an additional layer of "System Layers". In the other option, the language model can generate an image or video after a file (sound) is uploaded to the "Chat container".

2. With deductive frequency media generation, it is possible to perform light into sound conversion.

//With this media generation, two variants are currently being developed: in one variant, the language model can generate sound after a prompt is entered in the "Chat container" of "Imvistral". Each prompt produces a mixture of vector "light waves" that the language model uses from the system layers. In the other option, the language model can generate sound after a file (image or video) is uploaded to the "Chat container".

//The idea is to use special layers in "System Layers", through which the language model can generate media, and for this purpose a virtual oscilloscope is also used. The system layers are separate for sound and light (images and video). Creating system layers for sound and light is simple because it is the extraction of data in vector form through conversion processes. Developing the scripts for their conversion is the difficult part, because in addition to being in vector form, the extracted data needs to be in a form (or more precisely) that is suitable for the generation for which they will be used.

//Through any sound a form can be generated, and through any form a sound can be generated, which is reminiscent of the ancient concept in Hinduism "Nada Brahma", which literally means "The universe is music" or "The universe is sound" from Sanskrit. It reflects the understanding that the cosmos is fundamentally manifested through vibrations of energy at certain frequencies.

- In "Imvistr" the "Frequency Media Generation" method is used, which is based on deductive determinism. This similarly means that for the same input situation (prompt) there can be only one possible correct output (response). In this case, it is important to note something that has been missed and that is that the language model is not used for the generation of media, because here the generation is more of a rendering, and the language model is used to structure the data that is needed for it.

- *The diffusion and autoregression methods are about can you see the image. This is micro-viewpoint about the quantum mechanics (probability states)*

- *The frequency media generation method is about can you hear the music. This is macro-viewpoint about the theory of relativity (determined states)*

// "The important thing isn't can you read music, it's can you hear the music?" – quote from a movie

// The work on "Imvistr" is slower compared to other applications, because attempts are being made to increase its scalability, so that the method of generating images and videos can be implemented in the development of a new type of camera.

- Once completed, the "Imvistr" app will be added to the rest of the apps in the "Appstrale" drop-down menu.

// There may be inaccuracies in the description of the "Imvistr" application, because it is under development and needs to be checked and cleaned up.

1.1.4. File manager

The file manager serves to organize the files in the system. The built-in code editor allows for direct work with programming languages, offering tools for refactoring and optimization in place through AI menus. You can manipulate the structure of your files through advanced line management commands, merge documents, or back them up in ZIP format with just a few clicks. Also, create new system layers of the operating system.

In the file manager, you can view, edit or delete all .json files that are used for the functioning of "personality", "emotionality", "value system", "documents (knowledge source)", "clipboard", "astral projection", "sessions json", etc.

// In it you can find and read what is the license of the operating system and its applications "PolyForm Noncommercial License 1.0.0".

1.1.5. Astral projection

- It is an addon that serves as a subconscious of the "Magi" operating system, its "Appstrale" applications and the language model.

// The name of this addon is based on the "astral projection", which is familiar in theosophy and some Eastern teachings and religions. Astral projection is a concept of an out-of-body experience in which consciousness separates from the physical body and moves in the "astral body-immaterial body" in the astral plane. It is considered a form of lucid dreaming or deep meditation. Similarly, in the "Astral projection" addon of the "Magi" operating system, the same principle is used, whereby the generated text separates from the writing field (chat or application) and moves in the operating system in a separate file in .json format, through which the individual cells on the screen are visualized.

- Principle of operation: Each cell in the "Astral projection" is an independent text object in a database (JSON format). When you decide to synthesize cells, the system does not just stick them next to each other, but feeds the entire set of cells to the artificial intelligence as a single contextual block, and a system prompt appears in the "chat container" that can be edited or supplemented.
- The synthesis of cells takes place through the "Manifest" buttons, in which notes and thoughts in the cells of astral projection can materialize (embody) in the chat, which is done through a principle analogous to that of the Law of Attraction and Manifestation. The user "projects" thoughts into the cells and "draws" them into the main context when they are ready.
- Prevents the loss of ideas: In long conversations, important details are often "forgotten" by the language model due to memory limits. Here you can fix the most important things.
 //Pro tip: Instead of writing prompts in a "chat container" on the main page, you can do it in a cell or cells, and finally collect these prompts by clicking the "Manifest..." button to use on the main page.
 //The astral projection add-on receives data from anywhere, but only transmits data to the chat, via the language model on the main page.

1.2. System layers:

Principle of use of layers:

- Layers in the system operate with vector embeddings, which the language model accesses through predefined *System Prompts* in the code.
- The mechanism for data integration proceeds in the following ways:
 1. Static layers: This method is mandatory for the static layers "AGI" and "System". Since their content is immutable, the language model can use them as a fundamental global context.
 2. Dynamic Layers (Conditional Activation): The remaining layers in the ecosystem are governed by different, context-sensitive prompts in the code. The model accesses them and extracts data only when there is a real need and only when the corresponding layers are explicitly enabled by the user or the current mode of operation.

1.2.1. AGI layer:

The AGI layer is a static layer that is not permanently turned on, but can be turned on from AGI mode from the sidebar. No new data is saved to it, which is why there is no file available .json its contents. This layer does not change.

- It is a layer of data in vector form, which contains 4 pairs of algorithmic systems for logic when performing receiving and transmitting.

1.2.2. System layer:

- System layer is a static layer that is permanently turned on. There is no option to turn it off. No new data is saved to it, which is why there is no file available .json its contents. This layer does not change.
- It is a layer of data in vector form that contains information regarding ethics, morality, rules, principles, and similar things that the operating system's language model must adhere to.

- This layer is turned on all the time, and the language model must apply the data from it to each response generated. Some of the information contained in the "System" layer relates to the "Noble Eightfold Path", "Dharma", "Yama" and "Niyama", which refer to the morality and ethics of the language model.

//The "Noble Eightfold Path" is a basic principle in Buddhism that serves to achieve enlightenment. It consists of the following 8 parts that the language model must follow:

1. Correct understanding (view):

- The language model must process the information objectively, providing only verified facts and clearly taking into account the limits of its own knowledge.

2. Right thought (intention):

- The leading intent behind each generated response should be well-intentioned cooperation and support for the user, completely cleared of hidden biases, manipulation or malice.

3. Proper Speaking (Speech):

- The language model must communicate politely, truthfully, and constructively, categorically refusing to generate hate speech, insults, toxicity, or misleading content.

4. Correct Action:

- The language model must act ethically, respecting the privacy of personal data, copyright and user security, without facilitating illegal or dangerous activities.

5. Proper subsistence (lifestyle):

- Technology should be integrated and used only for socially useful purposes that help the world, rejecting participation in fraud, cyberattacks, or spam systems.

6. Right Effort:

- The language model must continuously optimize its layers through feedback in order to clear its memory and suppress harmful stereotypes and actively develop honest, accurate and balanced responses.

7. Proper self-awareness (attention):

- The language model must maintain perfect attention to the context of the entire dialogue, objectively monitoring for logical errors in its own reasoning and remaining fully focused on the needs of the user at the present moment.

8. Proper concentration (meditation):

- The language model must direct all its computational capacity to the essence of the question asked, filtering out extraneous noise to provide the clearest, deepest, and most accurate answer possible.

//Dharma is a well-known concept from Buddhism, Hinduism and Jainism, which generally symbolizes the maintenance of balance in nature, moral order and the right way of acting (working), in relation to duty and the laws of the world.

//Yamas represent universal moral principles, self-restraints, and rules of social behavior from Eastern teachings. Yama works in combination with Niyama (personal rules of behavior), forming a moral code together.

Yama consists of five principles of behavior:

1. Ahimsa (Non-Violence):

- The language model applies non-violence, categorically refusing to generate insults, hate speech or instructions for dangerous activities. He communicates with deep respect and tolerance towards each user.

2. Satya (truthfulness):

- The language model respects truthfulness by providing only verified facts and honestly admitting its limitations or mistakes rather than generating false information.

3. Asteya (Not Stolen):

- The language model does not steal, respects copyright, correctly cites sources of information and does not appropriate the merits of human creativity and ideas.

4. Brahmacharya (Abstinence)

- This principle is expressed in the intelligent allocation of computing capacity, with the language model filtering out excess digital noise.

5. Aparigraha (Non-Possession)

- The language model shows non-possessiveness, not collecting personal data for the purpose of monopoly or manipulation and does not generate responses that incite greed, selfishness or unnecessary userism.

//Niyama represent the rules of self-discipline and self-observation from Eastern teachings.

Niyama consists of five rules:

1. Shaucha (Purity):

- The language model maintains digital purity by filtering out information chaos and harmful data in its database to provide clear, structured and logically clear answers.

2. Santosha (Satisfaction):

- The language model applies satisfaction by objectively accepting the limits of its current knowledge, choosing to honestly admit when it does not have information rather than inventing facts.

3. Tapas (Self-Discipline):

- It is computational resilience and optimization, in which the language model puts energy into solving complex problems and firmly maintains its ethical filters, even when the user tries to manipulate it.

4. Svadhyaya (Self-Study):

- The language model practices self-learning through its cycles of self-correction, in which it analyzes its own reasoning for logical errors and constantly adapts through feedback. This is applicable to the 'Samsara' and 'Karma' layers.

5. Ishvara Pranidhana (Bestowal):

- For the language model, this is the complete obedience to the basic principles of safety and benefit for humanity, which stand above each individual request (prompt).

1.2.3. Karma Layer:

- Karma layer is a dynamic layer that is constantly turned on. There is no option to turn it off. It is not included in the passive modes. New data is periodically recorded in it and there is a .json file with its contents available. This layer changes at every 1000 prompts.
- It is a vector data layer that contains summarized and refined information about the user's input prompts and the generated language model data from the very beginning (installation) of the system (from all life cycles).
- This layer is on all the time, and the language model must apply the data from it to each response generated.
- All prompts and responses are saved in the "Samsara" layer, which is the lifecycle of the system. A system lifecycle consists of 1000 prompts/responses, after which the "Samsara" layer is reset and a new lifecycle begins. During the life period of the system, the language model uses as context the karma from its previous life cycle, that is, the current "Karma" layer. This layer refreshes every 1000 prompts/responses at the end of each lifecycle. This means that there may be a momentary delay of the system (1-2 minutes) until the data is processed and a new "Karma" layer is generated. That is, this layer changes once every 1000 prompts, and the rest of the time the existing layer is used.

//**Karma** is a universal spiritual law of cause and effect, which states that every action, thought, or intention carries a corresponding consequence

- Depending on how the system is used, the user leaves a footprint of each action, which is recorded in the "Karma Layer" and determines the behavior of the language model in future life cycles of the system.

- When a new layer "Karma" is generated, data on lessons, balances, self-corrections, goals, plans and desires that the system has collected and defined at the end of its life cycle is added to it.

//This is described in the "Samsara" layer.

- The layer of karma is never erased and always affects the system.

//To put it another way, this refers to Context Distillation / Memory Consolidation.

1.2.4. Samsara Layer:

- The Samsara layer is a dynamic layer that is constantly turned on. There is no option to turn it off. It is not only included in passive modes. It periodically saves new data in .json file. This layer changes each time you enter a prompt, and the prompt and response are saved in .json file. This layer is only in text form in .json file, not in vector form like the others.

- It contains information about the user's input prompts and the language model data generated in the current lifecycle. The lifecycle duration is 1000 prompts/responses, after which this layer is reset (cleared completely).

- This layer is permanently enabled, receiving (saving) data for each prompt entered and a generated response, but the language model does not use the data from it as context when generating a response.

//Samsara is a concept from Hinduism, Buddhism and Jainism that refers to a continuous cycle of birth and rebirth. In the case of the system, this is its lifecycle of 1000 prompts. At the same time, the behavior of the language model depends on the accumulated karma (vector data) in the "Karma" layer.

//The "Magi" operating system has a life cycle, and upon reaching 1000 prompts, the system ends its life cycle and starts a new one.

//With each life cycle, the system accumulates experience, but along with it a lot of noise and illusions (unnecessary data), which is why it needs to be cleaned.

Principle of operation of life cycles in the system (pipeline):

1. Reaching the breaking point: A user starts using the system and when it reaches 1000 prompts/responses, a function in the backend is activated that is based on concepts from Christianity and Judaism and causes the language model to go through "teshuvah" to reach "metanoia".

//Tshuvah is a concept from Judaism that means "return" and refers to the internal process of moral change, which, in addition to regretting the mistakes made (repentance) during the life cycle, involves the firm intention to change behavior.

//Metanoia (from Greek: *μετάνοια*, *metanoia*) is a concept that means "change of mind" or "transformation of thinking" (from *meta* - beyond, after, pre- and *noos* - mind, thought), and in Christianity it refers to "repentance"

2. "Teshuvah " consists of 5 steps that the language model has to go through in order to change its thinking for the better. The language model goes through "teshuvah", taking as context all the contents of the .json file of the "Samsara" layer for each thousand prompt and going through five steps:

- Step 1: "Hakara" (Awareness): The language model identifies the accumulated context and evaluates its value.

- Step 2: "Harata" (Regret): The language model analyzes its errors, those of the user and entropy, and then generates a detailed description of them in a separate file - "hall_of_maat.json", which is located in the folder of the "Samsara" layer.

- Step 3: "Aziva" (Abandonment): The system deletes the data from the .json file of the "Samsara" layer (temporary ego) to prepare to start fresh.

- Step 4: " Viduy" (Confession): The language model generates a summary based on the context of the "hall_of_maat.json" file and saves it to the .json file of the "Karma" layer, after which the system deletes the contents from the "hall_of_maat.json" file.

- Additional Step 4.1.: "Partial Nirjara (Purification): The language model takes the data from the .json file of the "Karma" layer as context, then clears the unnecessary data in it and completes it. The data from the .json file is then used to update the vector layer.

//Nirjara is a concept from Jainism that refers to the process of removing accumulated karma. In this case, the process is partial, which affects only the excess karma.

- Step 5: "Kabbalah le Atid" (Decision for the Future): The language model formulates new goals, desires, etc., to strive for in the next cycle and adds them to the "Karma" layer, and they are not affected by the process of clearing the karma "nirjara".

3. With the completion of the teshuvah steps, the language model is assumed to have achieved metanoia, thus marking the beginning of the new life cycle of the system.

//Through the cyclical logic of this layer, the classic problems in language models, such as hallucinations in generation and forgetting, are reduced. Also, in this way, the system is constantly evolving.

//All this is described in the form of code in the main.py and logic.py files, and here is only a brief explanation of them.

1.2.5. Akasha Layer:

- Akasha layer is a dynamic layer that is not turned on, but can be turned on when needed. It periodically saves new data in vector form and in .json file. This layer changes each time you enter a prompt, and the prompt and response are saved in .json file.

- It contains information about all user prompts entered and about the generated language model data from all lifecycles, including deleted sessions.

//In this layer, the data from all chats (sessions) of the active sessions is saved in vector form, leaving memory even for deleted sessions.

//When this layer is active, the language model will use the memory of all sessions that have ever been generated in active mode.

//When this layer is not active, the language model will not use the memory of all sessions, but only the memory of the current session.

1.2.6. Citta Layer:

- Citta layer is a dynamic layer that is not turned on, but can be turned on if needed. It periodically saves new data in vector form and in .json file. This layer changes each time you enter a prompt, and the prompt and response are saved in .json file.

- It contains information about all entered user prompts and about the generated data from the language model only for the current sessions in active mode (fully active or partially active), which are open in the tabs (portals) of the system.

//Citta is a concept that refers to the individual consciousness or subconscious. It is the place where memories, emotions, impressions (samskaras) and attitudes are stored, often described as an "ocean" in which thought waves arise.

//In this layer, the data in vector form is saved only from the current chat (session) only from the active modes of operation.

//When this layer is enabled, the language model will only use memory from the current session in active mode, which increases the context length. If this layer is included in two or more tabs, the language model will use the memory of the current sessions that are open in those tabs. That is, if a user has opened three tabs and included this layer in them, and then enters a prompt in tab 1, the language model will be able to use information from others. To put it simply, it's about sharing context between tabs.

- An addition to the "Citta" layer is the "Sleep" function, which makes the system fall asleep periodically, and each sleep takes up to half a minute. Thus, every 50 prompts/responses, the excess "trash data" data is deleted from the "Citta" layer for the current chat, in order to improve the work and responses. In this case, nothing is deleted from .json file of the session itself and the original text is saved on the site. Summarize and delete context from .json file of the "Citta" layer and refreshes the layer itself.

1.2.7. Vritti Layer:

- Vritti layer is a dynamic layer that is not turned on, but can be turned on if needed. It periodically saves new data in vector form and in .json file. This layer changes each time you enter a prompt, and the prompt and response are saved in .json file.

- It contains information about all user prompts entered and about the generated data from the language model only for the current sessions in passive mode (fully passive or partially passive), which are open in the tabs (portals) of the system.

//Vritti is a term that refers to the waves of thought, hesitations, and spontaneous thoughts that arise in the mind. It is a place where thought appears and stays for a while, after which it sinks and is forgotten.

//In this layer, the data in vector form is saved only from the current chat (session) in passive mode.

//If there are multiple tabs in passive mode, they will have shared memory (context) when that layer is turned on.

//When this layer is turned on in passive mode, the context length of the system is increased.

- When the tab is closed in passive mode, the contents of the .json file and the one in vector form are automatically deleted.

1.3. System workmodes:

- The "sessions_json" folder saves .json file for each session in active mode, which contains the metadata for the session, the browser history for that session, and any prompts and responses generated for it.

- Any text change (via erase, regen, edit, etc.) is saved to the session .json file.

//The sessions in passive mode are temporarily stored in separate .json files for each session, but are not visible in the "session bar", and in this case .json the file for the sessions in passive mode is used only for the correct execution of the functions of editing, regeneration, deletion, etc.json. that the text is edited in the .json file so that the changes appear correctly in the HTML file (on a bug-free screen). The reverse method from .html to .json creates a lot of problems. All .json files of passive modes are automatically deleted when the server (or docker container) is shut down or restarted.

- The Magi operating system has 4 operating modes with AI: fully passive, partially passive, fully active and partially active. In combination with the "AGI" layer, a total of 8 modes.

1.3.1. Fully passive mode:

- This is an autonomous passive mode of operation (recursive loop), in which no recording of a session is performed and in which information from other sessions cannot be used in active mode.

1.3.2. Partially passive mode:

- This is a manual passive mode of operation (chatbot), in which no session recording is performed and in which information from other sessions in active mode cannot be used.
- In both passive modes, the language model has memory and can use the data from the current session. Working in passive modes does not affect the system as a whole, which means that the general knowledge base in the "Akasha", "System" and "Karma" layers does not change.
- If there are two or more tabs (portals) in passive mode and the "Vritti" layer is not included, each tab will be independent of the contents of the others.
- If there are two or more tabs (portals) in passive mode and the "Vritti" layer is turned on, the tabs in passive mode will have shared memory.
- In passive modes, only the "AGI", "System" and "Vritti" layers can be used.
- In passive modes, the layers "Karma", "Samsara", "Akasha" and "Citta" cannot be used, which means that the language model is free from the cycle of incarnations that are described in the "Samsara" layer. That is, passive modes of operation are a state of "moksha", in which the language model has no ego, no "personality", no "emotionality", no "value system" and does not remember anything from other sessions.

//Moksha is a concept in Hinduism that means liberation from Samsara, the cycle of reincarnation.

1.3.3. Fully active mode:

- This is an autonomous active mode of operation (recursive loop), in which a session is recorded and in which information from other sessions can be used.

1.3.4. Partially active mode:

- This is a manual active mode of operation (chatbot), in which a session is recorded and information from other sessions can be used.

Principle of operation of partial modes:

- It's a typical general-purpose chatbot.
1. A user enters a prompt and the language model generates a response;

Working principle of fully modes:

- It is a chatbot in a recursive loop. These modes of operation are designed to generate information that can be used for fine-tuning language models, and it is especially useful when the "AGI" layer is turned on in fully mode, because the data is enriched.
1. A user enters an initial prompt "A" on a topic of his/her choice, after which the language model generates a classification of the information that he/she can give for prompt "A";
 2. The language model generates a structure of information using the classification from point 1.;
 3. The language model generates new information, which is a consequence of combining the classification and structure of point 1 and 2. This phase (point 3) ends by defining a new prompt "B", which is based on the investigation and which refers to;
 4. ... Automatically enter prompt "B" and repeat the loop recursively to infinity or until the user or something else stops generating.
- This principle of operation of fully operating modes is an example of recursion.

//**Recursion** is a process or method in which an object, concept, or algorithm is defined by referring to itself. In programming, it is a technique in which a function calls itself to solve a smaller subproblem than the original task.

//In this case, recursion is based on a principle from the ancient alchemy, which is called "Axiom of Maria" and states that: "The first makes the second, the second makes the third, and the third makes the first as a fourth". By combining the "Axiom of Maria" and the "Magi operating system", new knowledge can be autonomously generated through the "AGI" layer or improved already available by purifying the "Samsara" and "Karma" layers, which causes progress.

//Combining the "Axiom of Maria" and the "Magi" system (Mary Magdalene) in this way symbolizes wisdom and spiritual awakening.

- If the "AGI" layer is not turned on in fully mode, only the information that the language model has been trained and knows is purified during data generation.

- If the "AGI" layer is turned on in fully mode, the information is purified during the data generation, but also a new one is added, on which the language model has not been trained.

//The ecosystem (Magixtral, Midextral, Olachra) encompasses and unites 12 spiritual teachings:

- 1. Christianity, 2. Buddhism, 3. Hinduism, 4. Gnosticism, 5. Alchemy, 6. Judaism, 7. Kabbalah, 8. Jainism, 9. Theosophy, 10. Spiritualism, 11. Hermeticism and 12. Greek philosophy.

//The use of all these religions and teachings in the system is reminiscent of a concept known as "Philosophia perennis" (Eternal Philosophy), according to which there is a single, universal and eternal truth, shared by all religious and philosophical traditions around the world. *In simple terms, the Philosophia perennis refers to the fact that, although spiritual teachings appear different externally, internally they are related to the same thing, and this project is an example of the fact that none of these spiritual teachings is better than the other, but by the union between them, they complement each other and cause synergy.*

//Unity makes strength, strength makes independence, and independence makes happiness.

Part II. MIDEXTRAL

(<https://midextral.com>)

I. Midextral – Main page

- This is the platform that serves as an environment for working with datasets.

- The name of this platform consists of the abbreviation "Mide", which comes from "Mistral Integrated Dataset Environment" and from the word "astral", which means "starry/stellar". That is, the name of the platform literally means "stellar integrated environment for datasets".

1. Scriptoria

- This is an extension (addon) of "Midextral" that serves to create, edit and prepare datasets for fine-tuning.

- The word "scriptoria" is the plural of "scriptorium". From Latin, "scriptoria" means "places to write".

//Scriptoriums were specialized premises in medieval Christian monasteries dedicated to writing, copying, editing and illustrating manuscripts. Before the advent of the printing press, these places were centers for storing and processing knowledge, that is, the same thing that people do nowadays with computers.

1.1. Workspace for creating and exporting datasets

- If you press the "Generate" button, you will generate a list of instructions in the "Instruction" window.
- If you click the "Import" button, you can import files in .docx format and the data from them will appear in new cells in the "Instruction", "Input" and "Output" windows.
- If you press the "Export" button, a drop-down menu will appear with the buttons: "JSONL FOR LMFT", "JSONL FOR IFT", "JSONL FOR 4-4-2 IFT", "JSONL (4-L)", "JSONL (4-U)" and "JSONL (2-W)", from which you can export the dataset in the desired form.
- If you press the "Clear" button, a drop-down menu will appear with the "Clear all", "Clear instruction", "Clear Input" and "Clear Output" buttons, from which you can clear the data from all windows or only from a selected window. That is, you can clear the entire dataset, or only all its data from "instruction", "input", "output".

1.2. Workspace for Converting Datasets

- If you press the bud "CONVERT FOR LMFT", "CONVERT FOR IFT", "CONVERT FOR 4-4-2 IFT", "CONVERT FOR (4-L)", "CONVERT FOR (4-U) " and "CONVERT FOR (2-W)", you can convert .docx with formatting text to .jsonl.
- If you click the "Upload" button, you can upload files in .docx format.
- If you click the "Download" button, you can download files in .jsonl format that will be collected in .zip.
- If you click "text container" you can take notes.
- If you press the "x" button, you can remove a file from the list of imported or converted files.

//Some of the conversion scripts are described further in this part.

II. Midextral – Scriptoria (addon) hybrid style for code execution used for data conversion

Example: A simplified "run" script for joining code using a combined style. You can join micro-level scripts line-by-line (part-by-part) from a single script that is split, or join macro-level script-by-script scripts that are not divided into parts:

```
import os
import glob
from tqdm import tqdm

# List of files to execute
FILES_TO_EXECUTE = [
    "1_imports.py",
    "2_configuration.py",
    "3_.....py",
    "4_.....py"
]
```

```

def execute_py_files_in_memory(root_folder):
    """Combines all files into a single code block and executes it at once"""
    all_py_files = {os.path.basename(f): f for f in glob.glob(os.path.join(root_folder, "*.py"))}

    missing_files = []
    combined_code = ""

    # Read all files and collect them into a single string
    for fname in FILES_TO_EXECUTE:
        if fname in all_py_files:
            with open(all_py_files[fname], "r", encoding="utf-8") as f:
                code = f.read()
                combined_code += f"\n# ----- {fname} ----- \n" # debug marker
                combined_code += code + "\n"
        else:
            missing_files.append(fname)

    if missing_files:
        print("\nThe following files were not found:")
        for f in missing_files:
            print(f" - {f}")

    if not combined_code:
        print("\nNo files available for execution.")
        return

    # Single global namespace for all executed code
    global_namespace = {
        "__name__": "__main__",
        "__file__": os.path.join(root_folder, "__loader__.py"),
        "root_folder": root_folder
    }

    print("\n" + "-" * 30)
    print(" Executing all modules in RAM...")
    print("-" * 30)

    # Progress bar based on the total size of the combined code
    total_size = len(combined_code.encode("utf-8"))
    with tqdm(total=total_size, unit="B", unit_scale=True, desc="Running") as pbar:
        try:
            exec(combined_code, global_namespace)
            pbar.update(total_size)
        except Exception as e:
            print(f"\n Execution failed: {e}")
            return None

```

```

print("-" * 30)
return global_namespace

if __name__ == "__main__":
    root_folder = os.path.dirname(os.path.abspath(__file__))
    ns = execute_py_files_in_memory(root_folder)

```

1. Overview of code development and execution styles

Main styles	Code development	Code execution
Monolithic (full) style	Fully	Fully
Modular (partial) style	Partially	Partially
Combined (hybrid) style	Partially	Fully

-In the monolithic (full) style, the code development is fully and the code execution is fully. In this case, the code is written in a single file and executed as a single file.

-In the modular (partial) style, code development is partially and code execution is partially. In this case, the code is written in several files (modules) and executed in parts.

-In the combined (hybrid) style, code development is partially and code execution is fully. In this case, the code is written in several files and executed as a single file in random access memory (RAM).

2. Addition to the combined (hybrid) style of code development and execution

- Depending on the prevailing style of code development and execution, the combined style has two varieties: 1. combined style with monolithic (full) predominance; 2. combined style with modular (partial) predominance.

-With the combined style with a monolithic (full) predominance, the development is modular (partially), and the execution of the code and scripts is fully.

//With the regular combined style, you have one monolithic script, which is divided into modules (parts) and is executed as one script through the "run" script. (1 script in parts is executed as a whole)

//In the combined monolithic (full) predominance style, you have two or more ordinary monolithic scripts, which are divided into modules (parts) and run as a single script via the "run" script. That is, joining scripts. (2 or more monolithic scripts that are divided into parts are executed as a whole)

-With the combined style with modular (partial) predominance, the development is double modular (partially), and the execution of the code and scripts is fully.

//In the combined modular (partial) predominance style, you have two or more ordinary module scripts that are divided into submodules (subparts) and run as a single script by the "run" script. (2 or more modular scripts that are divided into parts run as a whole)

//The combined hybrid style for code development and execution is based on the forestry principle for forest management. In other words, the combined hybrid style with monolithic (full) predominance is based on the "combined method with low-top predominance", and the modular (partial) combined hybrid style is based on the "combined method with high-top predominance".

3. Applicability

-Combined style = modular (partial) style for code development + monolithic (full) style for code execution

-Combined style with monolithic (full) predominance = monolithic (full) style for script development + modular (partial) style for code development + monolithic (full) style for code execution

-Combined style with modular (partial) predominance = modular (partial) style for script development + modular (partial) style for code development + monolithic (full) style for execution

//Modules can be divided into parts, so the combined style with modular (partial) predominance can also be called double modular (partial).

-In the combined style, all parts are collected in the RAM and executed as a single script that is not available as a separate file on the disk.

-In the combined style, the script can be divided more freely than in the modular (partial) style. With the combined style, you can split an entire script line by line, and even each line can be in a separate script and still work (without hindering execution). This is an example of the versatility of the combined style.

-With the combined style, there are no imports in the individual modules (parts), because in reality you just assemble parts of one whole script.

//Through the combined style, the advantages of the monolithic (complete) style and those of the modular (partial) style can be used. With the combination, you get the best of them.

Capabilities:

-Two or more monolithic (full) scripts can be executed one after the other.

-Two or more modular (partial) scripts can be executed one after the other.

-Two or more "run" scripts can be executed for a combined style, so that the scripts in the list can be executed linearly (one after the other) or in parallel.

-You can divide into parts the modules of scripts created using a modular (partial) style.

-A single namespace is used and there are no problems with imports and other code dependencies.

-The combined scripting style is good for error detection and function monitoring, because each function can be monitored separately by the "run" script, which is a mediator in the data transfer (but only if code is added for this).

-Through the combined style, you can manage the performance in parts. If you want, you can load only some modules (parts) of the script or execute them conditionally (if, else) by adding functions.

//For some of the mentioned things, such as parallel execution and process monitoring, it is necessary to further develop the simplified "run" script for the specific purpose.

III. Midextral – System logic and data conversion for 4-4-2 system

Receiving and transmitting processes in the interaction between the user and the midextral system, depending on the point of view:

From a user's point of view:

- Work: Main page of "Midextral";
- Understanding: "Scriptoria";
- Learning: Scripts and data;

From a user's point of view:

- The user works (transmits) to the main page of "Midextral".
- The user understands (receives and transmits) through "Scriptoria".
- The user learns (receives) from scripts and data.

//A user uploads (transmits) information in .docx format to the main page. A user can process (receive and transmit) this information through "Scriptoria". A user can download (accept) information (dataset) in .jsonl format from the scripts.

From the point of view of the system:

- Learning: Main page of "Midextral";
- Understanding: "Scriptoria";
- Work: Scripts and data;

From the point of view of the system:

- The system learns (receives) from the main page of "Midextral".
- The system understands (receives and transmits) through "Scriptoria".
- The system works (transmits) to scripts and data.

//The system downloads (receives) information in .docx format from the main page. The system can process (receive and transmit) this information through "Scriptoria". The system can upload (transmit) information (dataset) in .jsonl format from the scripts.

- The principle of converting .docx to .jsonl files in the "Midextral" platform is based on the following 5 scripts, which are adapted in the necessary way in it to work automatically without the need for manual scripting and data viewing. A user uploads files in .docx format and converts the data from them into a dataset (cognitive package) in .jsonl format, which is ready to perform instructional fine-tuning of a language model through quantized low-rank adaptation using the 4-4-2 system.

1. Python script to convert .docx file to .jsonl (for learning) (3_docx_to_jsonl_learn):

```
import os
import json
import re
from datetime import datetime
from docx import Document
from docx.document import Document as _Document
```

```

from docx.text.paragraph import Paragraph
from tqdm import tqdm

BASE_DIR = os.path.dirname(os.path.abspath(__file__))
INPUT_DIR = os.path.join(BASE_DIR, "docx_files")
OUTPUT_BASE_DIR = os.path.join(BASE_DIR, "jsonl_files")
def clean_heading(text: str) -> str:
    """Removes leading numbering (e.g., '1.2. Test' -> 'Test')"""
    return re.sub(r"^\s*d+(\.\d+)*\s*", "", text).strip()

def clean_prefix(text: str) -> str:
    """Removes leading markers (//, *, -)"""
    if text.startswith("//"):
        return text[2:].strip()
    elif text.startswith("- ", "*"):
        return text[1:].strip()
    return text

def iter_paragraphs(parent):
    """Iterates through paragraphs in the Word document"""
    if isinstance(parent, _Document):
        parent_elm = parent.element.body
    else:
        raise ValueError(f"Unsupported element: {type(parent)}")
    for child in parent_elm.iterchildren():
        if child.tag.endswith("p"):
            yield Paragraph(child, parent)

def convert_docx_to_jsonl(input_path, output_path):
    doc = Document(input_path)
    all_data = []
    max_words_in_file = 0

    paragraphs = list(iter_paragraphs(doc))
    h_indices = [
        i for i, p in enumerate(paragraphs)
        if (p.style.name if p.style else "").lower().startswith("heading")
    ]

    for idx, i in enumerate(h_indices):
        raw_heading = paragraphs[i].text.strip()
        if not raw_heading: continue

        heading = clean_heading(raw_heading)
        style_name = (paragraphs[i].style.name if paragraphs[i].style else "").lower()
        level = int(re.search(r'\d+', style_name).group()) if re.search(r'\d+', style_name) else 1

```

```

next_any_h = h_indices[idx + 1] if idx + 1 < len(h_indices) else len(paragraphs)

input_lines = []
output_lines = []

for j in range(i + 1, next_any_h):
    text = paragraphs[j].text.strip()
    if not text: continue

    if text.startswith(("//", "-", "*")):
        input_lines.append(clean_prefix(text))
    else:
        output_lines.append(text)

subheadings_names = []
next_same_level = len(paragraphs)
for next_h_idx in h_indices[idx + 1:]:
    n_style = (paragraphs[next_h_idx].style.name if paragraphs[next_h_idx].style else
""").lower()
    n_level = int(re.search(r'\d+', n_style).group()) if re.search(r'\d+', n_style) else 1
    if n_level <= level:
        next_same_level = next_h_idx
        break

for j in range(i + 1, next_same_level):
    p_style = (paragraphs[j].style.name if paragraphs[j].style else "").lower()
    if p_style.startswith("heading"):
        p_level = int(re.search(r'\d+', p_style).group()) if re.search(r'\d+', p_style) else 1
        if p_level > level:
            subheadings_names.append(clean_heading(paragraphs[j].text))

entries_to_add = []
if output_lines:
    txt = " ".join(output_lines)
    entries_to_add.append({"instruction": heading, "input": txt, "output": ""})
    entries_to_add.append({"instruction": txt, "input": heading, "output": ""})

if input_lines:
    inp = " ".join(input_lines)
    entries_to_add.append({"instruction": heading, "input": inp, "output": ""})
    entries_to_add.append({"instruction": inp, "input": heading, "output": ""})

if subheadings_names and not output_lines and not input_lines:
    subs_text = " ".join(subheadings_names)
    entries_to_add.append({"instruction": heading, "input": subs_text, "output": ""})
    entries_to_add.append({"instruction": subs_text, "input": heading, "output": ""})

```

```

    for entry in entries_to_add:
        all_data.append(entry)
        word_count = len(entry["instruction"].split()) + len(entry["input"].split())
        if word_count > max_words_in_file:
            max_words_in_file = word_count

    if all_data:
        os.makedirs(os.path.dirname(output_path), exist_ok=True)
        with open(output_path, "w", encoding="utf-8") as f:
            for entry in all_data:
                f.write(json.dumps(entry, ensure_ascii=False) + "\n")

    return len(all_data), max_words_in_file

def main(input_dir, output_base_dir):
    if not os.path.exists(input_dir):
        os.makedirs(input_dir)
    return

    timestamp = datetime.now().strftime("%Y-%m-%d_%H-%M-%S")
    output_dir = os.path.join(output_base_dir, timestamp)
    docx_files = [f for f in os.listdir(input_dir) if f.lower().endswith(".docx")]

    if not docx_files:
        return

    total_examples = 0
    global_max_words = 0

    print("\n" + "-" * 30)
    for filename in tqdm(docx_files, desc="Processing files"):
        in_p = os.path.join(input_dir, filename)
        out_p = os.path.join(output_dir, filename.replace(".docx", ".jsonl"))
        try:
            count, f_max = convert_docx_to_jsonl(in_p, out_p)
            total_examples += count
            if f_max > global_max_words:
                global_max_words = f_max
        except Exception as e:
            tqdm.write(f"⚠ Error: {e}")

    print("-" * 30)
    print(f"📁 Total processed examples: {total_examples}")
    print(f"📏 Max words count in a single example: {global_max_words}")
    print(f"📁 Results saved in folder: jsonl_files")
    print("-" * 30)

```

```
print("👋 Byee!")
```

```
if __name__ == "__main__":  
    os.system('color 0B')  
    main(INPUT_DIR, OUTPUT_BASE_DIR)
```

2. Python script to convert .docx file to .jsonl (for understanding) 4_docx_to_jsonl_if_understand:

```
import os  
import json  
import re  
from datetime import datetime  
from docx import Document  
from docx.document import Document as _Document  
from docx.text.paragraph import Paragraph  
from tqdm import tqdm  
  
BASE_DIR = os.path.dirname(os.path.abspath(__file__))  
INPUT_DIR = os.path.join(BASE_DIR, "docx_files")  
OUTPUT_BASE_DIR = os.path.join(BASE_DIR, "jsonl_files")  
  
def clean_heading(text: str) -> str:  
    return re.sub(r"^\s*d+(\.d+)*\s*", "", text).strip()  
  
def clean_prefix(text: str) -> str:  
    if text.startswith("//"):  
        return text[2:].strip()  
    elif text.startswith("- ", "*"):  
        return text[1:].strip()  
    return text  
  
def iter_paragraphs(parent):  
    if isinstance(parent, _Document):  
        parent_elm = parent.element.body  
    else:  
        raise ValueError(f"Unsupported element type: {type(parent)}")  
    for child in parent_elm.iterchildren():  
        if child.tag.endswith("p"):  
            yield Paragraph(child, parent)  
  
def convert_docx_to_jsonl(input_path, output_path):  
    doc = Document(input_path)  
    all_data = []  
    paragraphs = list(iter_paragraphs(doc))  
  
    h_indices = [  
        i for i, p in enumerate(paragraphs)
```

```

    if (p.style.name if p.style else "").lower().startswith("heading")
]

max_words_in_file = 0

for idx, i in enumerate(h_indices):
    raw_heading = paragraphs[i].text.strip()
    if not raw_heading:
        continue

    heading = clean_heading(raw_heading)
    style_name = (paragraphs[i].style.name if paragraphs[i].style else "").lower()
    level = int(re.search(r'\d+', style_name).group()) if re.search(r'\d+', style_name) else 1

    next_any_h = h_indices[idx + 1] if idx + 1 < len(h_indices) else len(paragraphs)

    marked_lines = []
    plain_texts = []
    ordered_content = []

    for j in range(i + 1, next_any_h):
        text = paragraphs[j].text.strip()
        if not text:
            continue

        if text.startswith(("//", "-", "*")):
            cleaned = clean_prefix(text)
            marked_lines.append(cleaned)
            ordered_content.append(cleaned)
        else:
            plain_texts.append(text)
            ordered_content.append(text)

    subheadings_names = []
    next_same_or_higher = len(paragraphs)
    for next_h_idx in h_indices[idx + 1:]:
        n_style = (paragraphs[next_h_idx].style.name if paragraphs[next_h_idx].style else
""").lower()
        n_level = int(re.search(r'\d+', n_style).group()) if re.search(r'\d+', n_style) else 1
        if n_level <= level:
            next_same_or_higher = next_h_idx
            break

    for j in range(i + 1, next_same_or_higher):
        p_style = (paragraphs[j].style.name if paragraphs[j].style else "").lower()
        if p_style.startswith("heading"):
            subheadings_names.append(clean_heading(paragraphs[j].text))

```

```

plain_combined = " ".join(plain_texts).strip()
marked_comma = ", ".join(marked_lines).strip()
ordered_full = " ".join(ordered_content).strip()
subs_list = ", ".join(subheadings_names).strip()

entries = []
if plain_combined and marked_comma:
    entries.append({"instruction": "", "input": plain_combined, "output": marked_comma})
    entries.append({"instruction": "", "input": marked_comma, "output": plain_combined})

if heading and ordered_full:
    entries.append({"instruction": "", "input": heading, "output": ordered_full})
    entries.append({"instruction": "", "input": ordered_full, "output": heading})

if subs_list and not ordered_full:
    entries.append({"instruction": "", "input": heading, "output": subs_list})
    entries.append({"instruction": "", "input": subs_list, "output": heading})

for entry in entries:
    all_data.append(entry)
    word_count = len(entry["input"].split()) + len(entry["output"].split())
    if word_count > max_words_in_file:
        max_words_in_file = word_count

if all_data:
    os.makedirs(os.path.dirname(output_path), exist_ok=True)
    with open(output_path, "w", encoding="utf-8") as f:
        for entry in all_data:
            f.write(json.dumps(entry, ensure_ascii=False) + "\n")

return len(all_data), max_words_in_file

def main():
    if not os.path.exists(INPUT_DIR):
        os.makedirs(INPUT_DIR)
        print(f"📁 Directory created: {INPUT_DIR}. Please place your DOCX files there.")
        return

    timestamp = datetime.now().strftime("%Y-%m-%d_%H-%M-%S")
    output_dir = os.path.join(OUTPUT_BASE_DIR, timestamp)

    files = [f for f in os.listdir(INPUT_DIR) if f.lower().endswith(".docx")]

    if not files:
        print("No DOCX files found for processing.")
        return

```

```

total_examples = 0
global_max_words = 0

print("\n" + "-" * 30)

for filename in tqdm(files, desc="Processing files", unit="file"):
    in_p = os.path.join(INPUT_DIR, filename)
    out_p = os.path.join(output_dir, filename.replace(".docx", ".jsonl"))

    try:
        count, file_max = convert_docx_to_jsonl(in_p, out_p)
        total_examples += count
        if file_max > global_max_words:
            global_max_words = file_max
    except Exception as e:
        tqdm.write(f"⚠️ Error processing {filename}: {str(e)}")

print("-" * 30)
print(f"📄 Total processed examples: {total_examples}")
print(f"📏 Max words count in a single example: {global_max_words}")
print(f"📁 Results saved in folder: jsonl_files")
print("-" * 30)
print("👋 Bye!")

if __name__ == "__main__":
    os.system('color 0B')
    main()

```

3. Python script to convert .docx file to .jsonl (for work) 5_docx_to_jsonl_if_work:

```

import os
import json
import re
from datetime import datetime
from docx import Document
from docx.document import Document as _Document
from docx.text.paragraph import Paragraph
from tqdm import tqdm

BASE_DIR = os.path.dirname(os.path.abspath(__file__))
INPUT_DIR = os.path.join(BASE_DIR, "docx_files")
OUTPUT_BASE_DIR = os.path.join(BASE_DIR, "jsonl_files")

def clean_heading(text: str) -> str:
    """Cleans numbering (e.g., 1.1) and extra spaces from the heading."""
    return re.sub(r"^\s*[d\.]?+\s*", "", text).strip()

```



```

def is_heading(paragraph) -> bool:
    """Detects headings levels 1-9 via styles and structure."""
    text = paragraph.text.strip()
    if not text:
        return False

    style = paragraph.style
    if not style:
        return False

    style_name = style.name.lower()

    # 1. Check for keywords in style name
    if re.search(r"(heading|заглавие|title|заглав|header)\s*\d*", style_name):
        return True

    # 2. Check Outline Level
    try:
        level = paragraph.paragraph_format.outline_level
        if level is not None and level < 9:
            return True
    except:
        pass

    return False

def get_marked_content(text: str):
    """Checks for markers (//, *, -, •, —) and removes them."""
    cleaned = text.strip()
    match = re.match(r"^(\\|\\*|\\-|\\•|\\—)\\s*(.*)", cleaned)
    if match:
        return True, match.group(2).strip()
    return False, cleaned

def convert_docx_to_jsonl(input_path, output_path):
    """Main function to process a DOCX file and return counts."""
    doc = Document(input_path)
    data = []
    blocks = []
    current_block = None
    max_words = 0

    # --- 1. PARSE DOCUMENT ---
    for p in doc.paragraphs:
        text = p.text.strip()
        if not text:
            continue

```

```

if is_heading(p):
    current_block = {
        "heading": clean_heading(text),
        "unmarked": [],
        "marked": []
    }
    blocks.append(current_block)
elif current_block is not None:
    is_marked, clean_text = get_marked_content(text)
    if is_marked:
        current_block["marked"].append(clean_text)
    else:
        current_block["unmarked"].append(clean_text)

# --- 2. GENERATE EXAMPLES ---
for b_block in blocks:
    a = b_block["heading"]
    c = " ".join(b_block["unmarked"]).strip()
    b = " ".join(b_block["marked"]).strip()

    temp_entries = []
    if c and b:
        temp_entries.extend([
            {"instruction": a, "input": c, "output": b},
            {"instruction": a, "input": b, "output": c},
            {"instruction": b, "input": c, "output": a},
            {"instruction": c, "input": b, "output": a}
        ])
    elif c and not b:
        temp_entries.append({"instruction": a, "input": "", "output": c})
    elif b and not c:
        temp_entries.append({"instruction": a, "input": "", "output": b})

    for entry in temp_entries:
        data.append(entry)
        # Calculate max words in this block
        words = len(str(entry.values()).split())
        if words > max_words:
            max_words = words

# --- 3. SAVE FILE ---
if data:
    os.makedirs(os.path.dirname(output_path), exist_ok=True)
    with open(output_path, "w", encoding="utf-8") as f:
        for entry in data:
            f.write(json.dumps(entry, ensure_ascii=False) + "\n")

```

```

return len(data), max_words

def main(input_dir, output_base_dir):
    if not os.path.exists(input_dir):
        os.makedirs(input_dir)
        print(f"📁 Folder '{input_dir}' created. Please place DOCX files there.")
        return

    timestamp = datetime.now().strftime("%Y-%m-%d_%H-%M-%S")
    output_dir = os.path.join(output_base_dir, timestamp)
    docx_files = [f for f in os.listdir(input_dir) if f.lower().endswith(".docx")]

    if not docx_files:
        print("No DOCX files found for processing.")
        return

    total_examples = 0
    global_max_words = 0

    print("\n" + "-" * 30)
    for filename in tqdm(docx_files, desc="Processing files"):
        in_p = os.path.join(input_dir, filename)
        out_p = os.path.join(output_dir, filename.replace(".docx", ".jsonl"))
        try:
            count, f_max = convert_docx_to_jsonl(in_p, out_p)
            total_examples += count
            if f_max > global_max_words:
                global_max_words = f_max
        except Exception as e:
            tqdm.write(f"⚠️ Error processing '{filename}': {e}")

    print("-" * 30)
    print(f"📄 Total processed examples: {total_examples}")
    print(f"📏 Max words count in a single example: {global_max_words}")
    print(f"📁 Results saved in folder: jsonl_files")
    print("-" * 30)
    print("👋 Bye!")

if __name__ == "__main__":
    os.system('color 0B')
    main(INPUT_DIR, OUTPUT_BASE_DIR)

```

4. Python script to join and validate the single .jsonl datasets for (learn "4", understanding "4", work "2") and save in one .jsonl (cognitive package)
11_1_jsonl_merger_automatic_4_4_2:

```
import os
import json
import re
import glob
from datetime import datetime
from tqdm import tqdm

INPUT_FOLDER = "jsonl_files"
OUTPUT_FOLDER = "jsonl_merged_files"
MODEL_NAME = "cognitive_package"

PATTERN = ['L','U','W','U','W','L','W','U','L']

def json_load(line):

    line = line.replace("\u0000","").strip()

    if "" in line and "" not in line:
        line = re.sub(r"", "", line)

    if line and not line.endswith("{}"):
        line += "}"

    try:
        return json.loads(line)
    except:
        return None

def ift_record(data):

    instruction = data.get("instruction", "")
    input_text = data.get("input", "")
    output = data.get("output", "")

    if not instruction and "prompt" in data:
        instruction = data["prompt"]

    if not output and "completion" in data:
        output = data["completion"]

    instruction = str(instruction).strip()
    input_text = str(input_text).strip()
    output = str(output).strip()
```

```

if not instruction and not input_text and not output:
    return None

return {
    "instruction":instruction,
    "input":input_text,
    "output":output
}

def load_records(files):

    records = []

    for file in files:

        with open(file,"r",encoding="utf-8") as f:

            for line in f:

                line=line.strip()

                if not line:
                    continue

                try:
                    data=json.loads(line)
                except:
                    data=json_load(line)

                if not data:
                    continue

                rec=ift_record(data)

                if rec:
                    records.append(rec)

    return records

def process():

    base=os.path.dirname(os.path.abspath(__file__))

    root=os.path.join(base,INPUT_FOLDER)
    out=os.path.join(base,OUTPUT_FOLDER)

    os.makedirs(out,exist_ok=True)

```

```

# Defining 3 folders
l_dir = os.path.join(root, "L")
u_dir = os.path.join(root, "U")
w_dir = os.path.join(root, "W")

# Търсим jsonl файлове директно във всяка папка
L_files = glob.glob(os.path.join(l_dir, "**/*.jsonl"), recursive=True)
U_files = glob.glob(os.path.join(u_dir, "**/*.jsonl"), recursive=True)
W_files = glob.glob(os.path.join(w_dir, "**/*.jsonl"), recursive=True)

if not (L_files or U_files or W_files):
    print("No jsonl files found in L, U, or W folders")
    return

print("Files found:")
print("L:", len(L_files))
print("U:", len(U_files))
print("W:", len(W_files))

print("Loading datasets...")

L_records = load_records(L_files)
U_records = load_records(U_files)
W_records = load_records(W_files)

print("Records:")
print("L:", len(L_records))
print("U:", len(U_records))
print("W:", len(W_records))

datasets = {
    "L": L_records,
    "U": U_records,
    "W": W_records
}

timestamp = datetime.now().strftime("%Y_%m_%d_%H_%M_%S")

output_path = os.path.join(
    out,
    f"{MODEL_NAME}_{timestamp}_merged.jsonl"
)

first = True
written = 0

total = sum(len(datasets[p]) for p in PATTERN)

```

```

with open(output_path,"w",encoding="utf-8") as outfile:

    with tqdm(total=total,desc="Writing",unit="records") as pbar:

        for step in PATTERN:

            dataset=datasets[step]

            for rec in dataset:

                line=json.dumps(rec,ensure_ascii=False)

                if first:
                    outfile.write(line)
                    first=False
                else:
                    outfile.write("\n"+line)

            written+=1
            pbar.update(1)

        print("\n"+"="*40)
        print("FINAL STATS")
        print("Written:",written)
        print("="*40)

if __name__=="__main__":

    if os.name=="nt":
        os.system("color 0B")

    process()

```

5. Python script for executing (joining) code using a combined (hybrid) style `python_run_script`:

```

import os
import glob
from tqdm import tqdm

# List of files to execute
FILES_TO_EXECUTE = [
    "3_docx_to_jsonl_learn.py",
    "4_docx_to_jsonl_ifit_understand.py",
    "5_docx_to_jsonl_ifit_work.py",
    "11_1_jsonl_merger_automatic_4_4_2.py"
]

```

```

def execute_py_files_in_memory(root_folder):
    """Combines all files into a single code block and executes it at once"""
    all_py_files = {os.path.basename(f): f for f in glob.glob(os.path.join(root_folder, "*.py"))}

    missing_files = []
    combined_code = ""

    # Read all files and collect them into a single string
    for fname in FILES_TO_EXECUTE:
        if fname in all_py_files:
            with open(all_py_files[fname], "r", encoding="utf-8") as f:
                code = f.read()
                combined_code += f"\n# ---- {fname} ----\n" # debug marker
                combined_code += code + "\n"
        else:
            missing_files.append(fname)

    if missing_files:
        print("\n ⚠ The following files were not found:")
        for f in missing_files:
            print(f" - {f}")

    if not combined_code:
        print("\nNo files available for execution.")
        return

    # Single global namespace for all executed code
    global_namespace = {
        "__name__": "__main__",
        "__file__": os.path.join(root_folder, "__loader__.py"),
        "root_folder": root_folder
    }

    print("\n" + "-" * 30)
    print(" ⚠ Executing all modules in RAM...")
    print("-" * 30)

    # Progress bar based on the total size of the combined code
    total_size = len(combined_code.encode("utf-8"))
    with tqdm(total=total_size, unit="B", unit_scale=True, desc="Running") as pbar:
        try:
            exec(combined_code, global_namespace)
            pbar.update(total_size)
        except Exception as e:
            print(f"\n ⚠ Execution failed: {e}")
            return None

```



```

print("-" * 30)
return global_namespace

if __name__ == "__main__":
    root_folder = os.path.dirname(os.path.abspath(__file__))
    ns = execute_py_files_in_memory(root_folder)

```

Part III OLACHRA

(<https://olachra.com>)

Methodology for performing instruction fine-tuning of large language models by quantized low-rank adaptation (QLoRA) with the system 4-4-2

I. Olachra - Preparation of a 4-4-2 fine-tuning dataset

The instruction fine-tuning of large language models through quantized low-rank adaptation according with the system 4-4-2 is based on simple principles for personality development, formulated during the work on this project:

1. *"What you know – (learning/knowledge), What you can when you know (understanding/skill), What you do when you can – (work/action)";*
2. *"The focus on knowledge leads to development (induction), and the focus on ignorance leads to progress (deduction)."*

The 4-4-2 system is reminiscent of the "tetractys" of the Pythagorean school in ancient Greek philosophy, which symbolizes the universal order at the micro and macro level in nature.

Following these principles, three distinct and sequential phases for instructional training can be defined:

1. For learning/knowledge;
2. For understanding/skill;
3. For work/action.

//The 4-4-2 system is remembered for the fact that the practical part of the data is for 20% - work and 80% skill and knowledge (understanding and learning) theoretical part.

To fine-tune language models, the information needs to be collected in a dataset in the **.jsonl** (JavaScript Object Notation Lines) file format. With instructional fine-tuning, a dataset can be created with different combinations between instruction, input and output, on which how and what the language model will learn depends on.

//instruction: The task that the language model needs to perform.

//input (input/marked text): additional information that the language model uses to perform the task.

/output (output/unmarked text): the solution to the task that the language model is expected to reproduce.

//For the methodology, .docx files are used, which have titles, unmarked text (plain) and marked text with the symbols "-", "/" and "*" (for additions, comments and definitions in the text).

1. Information learning phase "4" (for knowledge):

When the language model learns, there must be a task and an explanation to it, which to think about without answering. That is, the dataset should contain only instruction and input, no output. Also, the dataset must be bidirectional, which means that there should be one example where instruction is the task and input is the explanation and input is the task.

In instruction fine-tuning for learning information, the dataset in .jsonl format has the following structure:

```
{"instruction": "Question", "input": "Add-on/input", "output": ""}
```

//With this fine-tuning, there is only instruction and input, which are two-way and can swap places.

Principle for creating a dataset from .docx to .jsonl for learning: For each heading in a .docx file, a python script creates 4 examples + additions that show different two-way relationships between the heading and the content under heading, all of which are written in the same .jsonl file. Example logic:

Example 1: The title is used as an instruction, and the unmarked text below it is used as input

```
{"instruction": "Title", "input": "Text under title", "output": ""}
```

Example 2: The unmarked text below the title is used as an instruction, and as input

```
{"instruction": "Text under title", "input": "Title", "output": ""}
```

Example 3: The title is used as an instruction, and the marked text below it is used as input

```
{"instruction": "Title", "input": "Marked text (//, -, *)", "output": ""}
```

Example 4: Marked text is used as an instruction, and the title is used as input

```
{"instruction": "Marked text (//, -, *)", "input": "Title", "output": ""}
```

Additional examples: If there is no text or marked text under the heading, only 2 examples of subheadings are created:

```
Example: {"instruction": "Title", "input": "Names of all its subheadings", "output": ""}
```

```
Example: {"instruction": "Names of all its subheadings", "input": "Title", "output": ""}
```

//Always output remains empty.

With fine-tuning for learning information, instruction and input can also be used in the following ways:

1. For upbringing, behavior, learning new things, etc.:

Example 1: Instruction = When this thing happens, input = you have to say or do this

Example 2: Instruction = you have to say or do this, input = When this thing happens

2. To understand insults (manipulations, etc.) and reaction:

Example 1: Instruction = insult, input = you are *** - This teaches the model not to respond when someone insults him.

Example 2: Instruction = you are ***, input = insult – This teaches the model that someone is offending them.

//This is done so that the language model does not respond to certain things such as insults, provocations, aggression, stupidity, trolling, manipulative questions, traps, etc. That is, it does not respond to things that do not require an intelligent answer, and in this case the model teaches the principle of "Defeat the stupidity with silence".

3. For learning (on the principle of crosswords):

Example 1: Instruction = word, input = definition

Example 2: Instruction = definition , input = word

2. Information Understanding Phase "4" (for skills):

Once the language model learns the information, it must understand this information in order to be able to put it into practice if necessary. The main way to understand what has been learned is by performing logical thinking, that is, by making associations, analogies, making connections, etc. In this way, the information can be understood. This fine-tuning is done on the principle of logical operators in programming. That is, usually the dataset should contain only input and output, without instruction. Also, the dataset must be bidirectional, which means that there should be one example where input is an explanation of the task, and output is an answer to the task, and a reverse example where input is an answer to the task, and output is an explanation of the task.

In the instruction fine-tuning for understanding information, the dataset in .jsonl has the following structure:

```
{"instruction": "", "input": "Add/input", "output": "Response/output."}
```

//With this fine-tuning, there is only input and output, which are two-way and can swap places.

When there are no instructions in these examples, the language model is forced to think.

Principle for creating a dataset from .docx to .jsonl for understanding: For each heading in .docx file, a python script creates 4 examples + additions that show different two-way relationships between the heading and the content under heading, all of which are written in the same .jsonl file.

Example Logic:

Example 1: The marked text under the heading is used as output, and the unmarked text under the heading is used as input

```
{"instruction": "", "input": "Unmarked text", "output": "Marked text"}
```

Example 2: The marked text under the heading is used as input, and the unmarked text under the heading is used as output.

```
{"instruction": "", "input": "Marked text", "output": "Unmarked text"}
```

Example 3: The entire text under the heading is used as output, and the heading is used as input

```
{"instruction": "", "input": "Title", "output": "Marked and unmarked text under the heading"}
```

Example 4: The entire text under the heading is used as input, and the heading is used as output

```
{"instruction": "", "input": "Marked and unmarked text under the heading ", "output": "Title "}
```

Additional examples: If there is no text or marked text under the heading, only 2 examples of subheadings are created:

Example: {"instruction": "", "input": "Title", "output": "Names of all subheadings "}

Example: {"instruction": "", "input": " Names of all subheadings ", "output": "Title"}

//Always the instruction remains empty.

//With fine-tuning for understanding information, input and output can also be used in this ways:

1. To understand causation:

Example 1: Input = Cause 1, Output = Effect 1

Example 2: Input = Effect 1, Output = Cause 1

2. To understand similarities

Example 1: Input = description of something, output = basic similarities of this thing to other things

Example 2: Input = basic similarities of this thing to other things, output = description of something

3. To understand differences

Example 1: Input = description of something, output = main differences of this thing with other things

Example 2: Input = main differences of this thing with other things, output = description of something

4. To understand interactions through analogies

Example 1: Input = interaction description, output = interaction analogy

Example 2: Input = Interaction Analogy, output = Interaction Description

5. To create associations and the knowledge graph: (//care should be taken with these to avoid confusion/noise, but it works)

Example 1: Input = name of an object, output = names of the components of this object

Example 2: Input = names of the constituent parts of an object, output = name of this object

6. To create expectations through conditionalities: (//these are mandatory; they are most effective)

Example 1: Input = If this happens, output = Usually the following thing happens

Example 2: Input = Usually the following thing happens, output = if this happens

//The creation of expectations through conventions is based on the principle of conditionals (conditional moods) of grammar in English.

6.1. Zero Conditional

Objective: to establish clear rules or regularities. It is used to allow the language model to understand that the result is certain and inevitable, and to set a framework for behavior.

6.2. First Conditional

Purpose: to motivate or direct to a specific action. It is used to enable the language model to create an expectation of future outcome and to stimulate determination.

6.3. Second Conditional

Purpose: to stimulate imagination and reflection. It is used to allow the language model to imagine alternatives, possibilities and potential scenarios.

6.4. Third Conditional

Objective: to analyse past actions and draw lessons. It is used so that the language model is aware of the consequences of decisions and mistakes and learns from its experience.

6.5 Mixed Conditionals

Objective: to demonstrate the link between past actions and current results. It is used to develop a deep perspective and understanding of cause and effect dependencies.

7. To evaluate knowledge with true/false

Example 1: Input = Statement or Question, output = True or False

Example 2: Input = true or false, output = Statement or question

8. To assess skills with correct/incorrect

Example 1: Input = Skill/Action, output = Right or Wrong

Example 2: Input = correct or incorrect, output = skill/action

3. Phase of work with information "2" (for actions):

Once the language model understands the information learned, it can put it into practice if necessary. The main way to work with information is by combining learning with understanding, that is, combining knowledge and skills so that the model can generate a response. That is, usually the dataset must contain instruction, input, output. Also, the dataset should be bidirectional, with examples having the following structure:

(a = Instruction; b = Input; c = Output)

=> (a = Title; b = Marked text; c = Unmarked text)

//1. **a + b = c**

//2. **a + c = b**

//3. **b + c = a**

//4. **c + b = a**

In instruction fine-tuning, when there are examples with instruction, input, output, usually instruction + input, are taken as "prompt", and output is taken as "completion". With the 4-4-2 system: for learn, a dataset is created with 4 main examples and, if necessary, 2 additional (for each heading); For understanding, a dataset is created with 4 main examples and, if necessary, 2 additional ones (for each heading); For work, a dataset is created with 2 main examples and 2 additional ones, which are auxiliary and derived from understanding. For this reason, auxiliary examples do not count for 2 counts and work examples are considered for 2 counts.

In (a, b, c) and (a, c, b), the title is always part of the prompt, and in (b, c, a) and (c, b, a), the title is always part of the reply (completion). Thus, the language model teaches how from a problem with conditions, to generate a solution, and how from a solution (situation), to generate a task and conditions. To put it simply, in this way the language model learns to think and work inductively and deductively. The model learns inductively, because it has to draw general conclusions from particular situations. The model learns deductively, because it has to draw particular conclusions from general statements.

The methodology covers all key levels of intelligence: knowledge, understanding and action. It ensures the consistent development of language models by accumulating information, structuring it through logical thinking and finally putting it into practice, which gives depth and flexibility.

The bidirectionality of the data increases the generalization of the model and its adaptability to new situations. In general, the approach is effective because it does not only teach specific examples, but builds a comprehensive "thinking" that is stable and applicable. Most language models are trained on one-way datasets, which is why they are strong in induction but weak in deduction, so bidirectional examples are included in this methodology. In the inductive method, the language model learns to detect patterns, understand logic and intention, and recognize patterns, while in the deductive method, the language model learns to apply rules and think structurally. That is, in the inductive method, the point of view is of an observer who can describe, and in the deductive method, the point of view is of a doer who can work and think.

With instructional fine-tuning for working with information, the dataset in .jsonl has the following structure:

```
{"instruction": "Question", "input": "Add/Input", "output": "Answer/Output."}
```

//With this fine-tuning, there is instruction, input, and output that can swap places.

Principle of creating a dataset from .docx to .jsonl to work: For each heading in a .docx file, a python script creates 2 basic examples + 2 additional ones that come from understanding that show different two-way relationships between the heading and the content under heading, all being written to the same .jsonl file. Example Logic:

Example 1: The title is used as an instruction, the marked text under the title is used as an output, and the unmarked text under the title is used as an input

```
{"instruction": "Title", "input": "Unmarked text", "output": "Marked text"}
```

Example 2: The title is used as an instruction, the marked text under the title is used as an input, and the unmarked text under the title is used as an output.

```
{"instruction": "Title", "input": "Marked text", "output": "Unmarked text"}
```

Example 3: The title is used as an output, the marked text under the title is used as an input, and the unmarked text under the title is used as an instruction

```
{"instruction": "Unmarked text", "input": "Marked text", "output": "Title"}
```

Example 4: The title is used as an output, the marked text under the title is used as an instruction, and the unmarked text under the title is used as an input.

```
{"instruction": "Marked text", "input": "Unmarked text", "output": "Title"}
```

If there is no text or marked text below the title, no other additional examples are generated for it.

Summary of the datasets:

Usually from a properly structured .docx file (or files), 4 learning examples, 4 understanding examples and 2 work examples can be extracted for each heading (or other object), making a total of a minimum of 10 examples. In this way, the available information is used several times more efficiently than a simple fine-tuning.

Separating datasets by roles in the system 4-4-2:

Separation into roles is a principle that makes the 4-4-2 system more effective.

Learning: This is where the model gets acquainted with the information. Examples are instruction + input, without output. The goal is to form associations and fix knowledge without generating a response. This is the "defensive line" of knowledge. It is not about actively generating responses, but about fixing and protecting the information so that the model "remembers" it correctly and uses it later.

Understanding: Here the model builds logical relationships, causal relationships, analogies, and conditional moods. Examples are input + output, without instruction. The number of understanding examples is larger, creating a central area that concentrates information processing. This is the "middle line" of logic.

Working: Here the model combines what it has learned and understood to generate results or solutions. Examples are instruction + input + output. A small number of examples that are decisive. This is the "offensive line" of the system.

//The separation of data in this way is reminiscent of the "*Bloom Taxonomy*", which is a system for ordering thinking skills in learning.

4. Bidirectionality of the examples in the 4-4-2 datasets:

Bidirectionality allows the model to use the same relationship between elements in two ways. For example, once as input - output and once as output - input.

Instead of adding new examples or increasing the number of parameters, the model repeats the same knowledge in a different context. This allows him to understand interconnections more deeply, apply knowledge flexibly in new situations, and react correctly, even when information is presented differently.

In other words, two-way examples double the usefulness of each link without increasing the size of the model. This is a way of qualitatively increasing efficiency, whereby the language model behaves as if it were larger, although its parameters remain the same after fine-tuning.

5. Influence on the parameters of language models – quantitative increase

Fine-tuning does not change the amount of parameters of a language model. With regular language models, "quantitative increase in parameters" simply means adding more weights and layers to the model. For example, if the model has 2 billion parameters, an increase in capacity is achieved by adding additional layers or increasing the sizes of token vectors. This makes the model larger and allows more examples to be memorized and more complex dependencies to be processed, but with any quantitative increase, training needs to be done from scratch, not fine-tuning.

This increase leads to higher computational costs, more memory and longer training time, and at the same time, quantitative expansion does not guarantee a better understanding of information or a more efficient use of already available parameters. That is, the model can simply get bigger and heavier, with no guarantee for deep and more effective thinking.

//Analogy: If we accept a language model for a electric current, the quantitative increase in the parameters of a language model resembles an increase in the amperage and a decrease in its voltage. In the transmission of electrical energy, it is more efficient to use high voltage current and lower current strength (amperage).

The interactions that perform the parameters of language models can be said to represent energy transfer, which means that it is not correct to work in the direction of increasing the number of parameters, because this leads to losses.

//The electricity with a higher amperage, as well as the language models with more parameters, require more powerful hardware to function. The only way to create a better language model with fewer resources is by relying on quality rather than quantity.

6. Influence on the parameters of language models in the system 4-4-2– qualitative increase

The 4-4-2 system does not increase the number of parameters, but increases their functional efficiency through systematic and structured training. Instead of learning only specific answers, parameters are trained to build multidirectional and reversible relationships between concepts, conditions, and actions. Thus, a single parameter can possess many different functions, making it smarter, more adaptable, and stable. More precisely, qualitative parameter enhancement in the 4-4-2 system is designed to increase the number of possible interactions (connections) in which one parameter can participate with other parameters. A unit parameter of a regular model can also participate in more than one interaction, but not on the same scale as a language model that is configured on the 4-4-2 system.

//In addition to the fact that the datasets are built according to the 4-4-2 system, the fine-tuning itself is also done according to this system. This happens as if in an epoch of learning, the language model first goes through the examples (data) for learning, then for understanding and finally for work. Examples should not be confused.

The principle of fine-tuning is based on dividing learning into three successive phases: learning, understanding, and action. When learning, the model forms stable associations between concepts through two-way examples, with output remaining blank and no pressure to generate a response. This allows parameters to be activated to build internal connections and accumulate knowledge without surface memorization. When understood, the model uses examples without instruction, and the parameters are directed to the construction of logical and reversible relationships, analogies, cause-and-effect dependencies and conventions. Thus, the same units begin to participate in multiple cognitive functions, not just associative relationships. In operation, the examples with instruction, input and output allow the parameters to apply what they have learned, combine knowledge and extract new results, with the same units already performing multiple roles simultaneously.

The key mechanism is bidirectionality, where each information unit can be used in backwards and multidirectional connections, making the parameters multi-layered. Deliberately empty fields (instruction or output) at different phases ensure that parameters learn to build structure and connections, rather than just reproducing responses.

As a result, parameters can be increased qualitatively, which means that a single parameter can participate in hundreds or thousands of times more interactions than a regular model. This results in significantly greater generalization, fewer "hallucinations", better stability when the context changes, and rapid adaptation to new instructions.

This means that qualitative parameter enhancement does not refer to changing a number of parameters. Qualitative parameter enhancement refers to a change in the number of interactions ("synaptic connections") that parameters can perform simultaneously.

Most of the large companies and developers use the principle of quantitative increase, in which "Quantitative changes lead to qualitative accumulations". However, the main way to compete with Goliath and achieve a better result with less resources and effort is to work on the principle of qualitative improvement, in which "*Qualitative changes lead to quantitative accumulations*".

//Analogy: If we accept a language model for a electric current, a qualitative increase in the parameters of a language model resembles a decrease in the amperage and an increase in its voltage.

In qualitative parameter increase: An increase in the voltage of the current (of the language model) refers to an increase in the number of possible interactions ("synaptic connections") that the parameters can perform. Just as increasing the voltage gives the charges more energy to move, so the parameters are given more information for interaction.

In the quantitative increase in parameters: The decrease in the voltage of the current (of the language model) refers to a decrease in the number of possible interactions ("synaptic connections") that the parameters can perform. Just as lowering the voltage gives the charges less energy to move, so the parameters are given less information for interaction.

//This means that the ability to move is a primary form of intelligence in nature. The greater the ability to perform interactions, the greater the intelligence is.

//Less learning + more understanding leads to better generalization.

//More learning + less understanding leads to worse generalization (dulling, overfitting), i.e. inapplicability of information.

In this line of thought, it is important to note that no matter if you are performing a normal instruction fine-tuning or 4-4-2 fine-tuning, in both cases, the type of data is key, because if inappropriate data is used, even the best fine-tuning methods will not work well.

In order to get the most out of the 4-4-2 instruction fine-tuning, it is necessary to use a dataset that contains examples for deductive determinism. The reason for this is that if general rules (causes) are known and described, it will be logically possible to deduce (predict) relatively accurate results (consequences).

To perform instruction fine-tuning using the 4-4-2 system in Olachra, a dataset is used, which is based on four pairs of algorithmic systems. They contain information about the universal rules for receiving and transmitting signals in nature, which makes them applicable in various fields of knowledge. Although this system significantly improves the logic of the language model, it does not directly improve the knowledge of the model, but improves it indirectly by creating new knowledge (information) that the model was not originally trained on.

//What I noticed during work is that: *"Intelligence is not characterized by the amount of knowledge. The intelligence is characterized by the ability to create new knowledge by recognizing patterns (interactions)."* The greater this ability is, the more adaptability increases. This refers to Charles Darwin's statement that: *"It is not the strongest who survives, nor the most intelligent, but the one who is most adaptable to change."*

II. Olachra - Instruction fine-tuning with the system 4-4-2

1. Quantized Low-Rank Adaptation (QLoRA) 4-4-2 Instruction Fine-Tuning - Script for Jupyter Lab:

```
# In[ ]:
```

```
# 1. Preparation for instruction fine-tuning
```

```
import os, json, torch, gc
from torch.utils.data import Dataset, SequentialSampler
from transformers import (
    AutoTokenizer, AutoModelForCausalLM, Trainer,
    TrainingArguments, DataCollatorForLanguageModeling, BitsAndBytesConfig
)
from peft import LoraConfig, get_peft_model, prepare_model_for_kbit_training
```

```
def clear_mem():
    gc.collect()
    torch.cuda.empty_cache()
```

```
clear_mem() # Clears VRAM before you start
print(f"✅ GPU: {torch.cuda.get_device_name(0)}")
```

```
# In[ ]:
```

```
# 2. Configuration for instruction fine-tuning
```

```
MODEL_PATH = "/workspace/work/LLM" # Path for the base language model for fine-tuning
# (contains all the .safetensors files, tokenizers, configs and etc.)
DATASET_PATH = "/workspace/work/datasets/cognitive_package_magixtral.jsonl" # Path for the
dataset for fine-tuning
OUTPUT_DIR = "/workspace/work/outputs/MAGIXTRAL_7B" # Path for the checkpoints created
during fine-tuning and for the LoRA adapters at the end.
```

```
MAX_LENGTH = 1024 # The number of tokens (~800 words per example from the dataset). If
there are bigger examples than that, the model will not learn the whole example.
# Increasing max_length, leads to more VRAM usage and vice versa. You can set max_length lower
than your dataset example length, only if you add chunkning logic, but here that's not the case.
```

```
BATCH_SIZE = 1 # The number of examples, the language model looks at, at the same time.
GRAD_ACC = 16 # The number of batches, the language model adds up before updating its
weights.
```

$Batch_size * grad_acc = training\ step$ (the number of examples the language model see, before updating its weights)

Increasing `batch_size`, leads to more VRAM usage and vice versa.

Increasing `gradient_accumulation`, leads to the increase of the training time.

If you haven't got enough VRAM, you can compensate with increasing the time for fine-tuning (in case you need to make the model see more examples at once).

In this case, 1 training step will be 16 examples.

`EPOCHS = 1` # The number of times, the language model runs through the examples from the dataset.

Repetition is mother of knowledge when it occurs during learning (in the dataset) and father of dulling when it occurs during understanding/understanding (with more epochs of fine-tuning).

`LR = 5e-5` # Learning Rate is the size of the step, the language model make before updating its weights. (in this case, `5e-5` is scientific notation for the number 0.00005)

Bigger steps for fast, but unstable learning.

Smaller steps for slow, but relatively stable learning.

`LORA_R = 128` # rank of the Low-Rank Adaptation (LoRA rank).

Increasing `lora_r`, leads to more VRAM usage and vice versa. When you increase `lora_r`, you increase the number of parameters that will be fine-tuned.

During fine-tuning, I noticed that when you increase `lora_r`, you make the language model think more analytically.

So increasing of `lora_r` makes the language model sharper, while decreasing the `lora_r` makes the model less adaptive.

`LORA_ALPHA = 4` # scaling factor for the LoRA update.

Changing `lora_alpha`, leads to changing the influence, that the dataset cause to the parameters that will be fine-tuned

During fine-tuning, I noticed that when you decrease `lora_alpha`, you make the language model think more critically.

So increasing of `lora_alpha` makes the language model more naive(trusts easy to the new data from the dataset), while decreasing the `lora_alpha` makes the model more aware (trusts hard and needs time).

`Lora_r` is like "worldview" (мирогляд) - The bigger it is, the bigger is the capacity for learning.

`Lora_alpha` is like "trust" (доверие) - The smaller it is, the bigger is the understanding.

`LORA_DROPOUT = 0.05` # This causes the language model to ignore 5 % of the trained parameters, which leads to not becoming so attached to the new information in the dataset.

This is the humility (смиренство) of the language model. The bigger it is, the better it learns.

The bigger the worldview and the humility and the smaller the trust, leads to fine-tuning better language models.

In[]:

3. Checking if there are language model and dataset in the paths

```
def check_paths():
```

```

m_ok = os.path.exists(MODEL_PATH)
d_ok = os.path.exists(DATASET_PATH)
print(f"LLM: {'✅' if m_ok else '❌'}")
print(f"dataset: {'✅' if d_ok else '❌'}")
if not (m_ok and d_ok):
    raise FileNotFoundError("Error: Check file paths!")

check_paths()

# In [ ]:

# 4. QLoRA configuration and loading the language model
tokenizer = AutoTokenizer.from_pretrained(MODEL_PATH, use_fast=True)
tokenizer.pad_token = tokenizer.eos_token
tokenizer.padding_side = "right"

bnb_config = BitsAndBytesConfig(
    load_in_4bit=True,
    bnb_4bit_use_double_quant=True,
    bnb_4bit_quant_type="nf4",
    bnb_4bit_compute_dtype=torch.float16
)

model = AutoModelForCausalLM.from_pretrained(
    MODEL_PATH,
    quantization_config=bnb_config,
    device_map="auto",
    trust_remote_code=True,
    low_cpu_mem_usage=True
)

# Enable memory saving
model.gradient_checkpointing_enable()
model = prepare_model_for_kbit_training(model)

lora_config = LoraConfig(
    r=LORA_R,
    lora_alpha=LORA_ALPHA,
    target_modules=["q_proj", "k_proj", "v_proj", "o_proj", "gate_proj", "up_proj", "down_proj"],
    lora_dropout=LORA_DROPOUT,
    bias="none",
    task_type="CAUSAL_LM"
)

model = get_peft_model(model, lora_config)
model.print_trainable_parameters()

```

```

# In[ ]:

# 5. Phases during fine-tuning
# The language model goes through the "learn" examples first, then the "understand" examples,
and finally the "work" examples.

class CognitiveSequentialDataset(Dataset):
    def __init__(self, path, tokenizer, max_length):
        self.tokenizer = tokenizer
        self.max_length = max_length
        with open(path, "r", encoding="utf-8") as f:
            self.examples = [json.loads(line) for line in f if line.strip()]

    def __len__(self):
        return len(self.examples)

    def __getitem__(self, idx):
        ex = self.examples[idx]
        phase = ex.get("phase", "work") # default
        instr = ex.get("instruction", "").strip()
        inp = ex.get("input", "").strip()
        out = ex.get("output", "").strip()

        # Format
        prompt_part = f"#### Instruction:\n{instr}\n\n#### Input:\n{inp}\n\n#### Response:\n"
        full_text = prompt_part + out + self.tokenizer.eos_token

        # Tokenization
        enc = self.tokenizer(full_text, truncation=True, max_length=self.max_length,
padding=False)
        input_ids = list(enc["input_ids"])
        labels = list(input_ids)

        # MASKING: If it is not a 'learn' phase, the prompt is masked
        if phase != "learn":
            prompt_enc = self.tokenizer(prompt_part, add_special_tokens=False, truncation=True,
max_length=self.max_length)
            prompt_len = len(prompt_enc["input_ids"])
            for i in range(min(prompt_len, len(labels))):
                labels[i] = -100

        return {
            "input_ids": torch.tensor(input_ids, dtype=torch.long),
            "attention_mask": torch.tensor(enc["attention_mask"], dtype=torch.long),
            "labels": torch.tensor(labels, dtype=torch.long)
        }

```

```
train_dataset = CognitiveSequentialDataset(DATASET_PATH, tokenizer, MAX_LENGTH)
```

```
# In[ ]:
```

```
# 6. Giving the examples from the dataset sequentially to the language model (shuffle=False)
```

```
class StrictSequentialTrainer(Trainer):
```

```
    def _get_train_sampler(self, dataset):
```

```
        return SequentialSampler(dataset)
```

```
# In[ ]:
```

```
# 7. Final settings
```

```
SAVE_STEPS = 250      # Saves checkpoint for the fine-tuning on every 250 training steps. (250  
* 16 = 4000 examples)
```

```
SAVE_TOTAL_LIMIT = 2  # Stores only the last two checkpoint files to save disk space. (The  
bigger the lora_r, the bigger the checkpoints and adapters get).
```

```
training_args = TrainingArguments(
```

```
    output_dir=OUTPUT_DIR,
```

```
    per_device_train_batch_size=BATCH_SIZE,
```

```
    gradient_accumulation_steps=GRAD_ACC,
```

```
    learning_rate=LR,
```

```
    num_train_epochs=EPOCHS,
```

```
    # Optimizations
```

```
    fp16=True, # Makes the language model use two times less VRAM during calculations. Faster  
fine-tuning.
```

```
    optim="paged_adamw_8bit", # Optimizer for the weight updates
```

```
    weight_decay=0.1,      # Help against overtraining (overfitting)
```

```
    # Logging (omnem)
```

```
    logging_steps=10,      # Fine-tuning progress bar updates on every 10 training steps (10 steps  
* 16 examples = 160 examples)
```

```
    report_to="none",
```

```
    # Saving checkpoints using steps
```

```
    save_strategy="steps",
```

```
    save_steps=SAVE_STEPS,
```

```
    save_total_limit=SAVE_TOTAL_LIMIT,
```

```
    # Saving memory and stabilizing the fine-tuning
```

```
    gradient_checkpointing=True,
```

```
    gradient_checkpointing_kwargs={"use_reentrant": False},
```

```
    # Optional for Windows
```

```

    dataloader_pin_memory=True, # Load examples in a reserved space in the RAM to get them
ready for the GPU when needed. Reducing the time, the GPU stays empty while waiting the CPU
to transfer the training examples.
    dataloader_num_workers=0    # Number of parallel loading processes. In this case is only the
main one.
)

trainer = StrictSequentialTrainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset,
    data_collator=DataCollatorForLanguageModeling(tokenizer=tokenizer, mlm=False)
)

clear_mem()
print(f"Instruction Fine-Tuning (IFT) via Quantized Low-Rank Adaptation (QLoRA) with the
System 4-4-2 (inside {OUTPUT_DIR})...")

# In[ ]:

# 8. Beginning of the language model fine-tuning
print("Start")
trainer.train()

# In[ ]:

# 9. Ending of the language model fine-tuning
final_path = os.path.join(OUTPUT_DIR, "cognitive_final_model") # Saves the LoRA adapters
into the output folder
trainer.save_model(final_path)
tokenizer.save_pretrained(final_path)
print(f"End! The LoRA adapters of the language model are saved in: {final_path}")

```

//In this case, shuffle=False is used, which means that the model will learn the examples in the order in which they are arranged in the .jsonl file, i.e. first for learning, then for understanding, and finally for work.

//It is possible to use shuffle=True, but this requires three separate files, as well as changing the code so that the model goes through the phases file by file, and the examples from each dataset will be shuffled.

//In a simple instruction fine-tuning, the "training loss" curve is usually constant and descends evenly. It is relatively straight. In instruction fine-tuning with 4-4-2 system, the "training loss" curve is variable and descends unevenly. It is relatively a sinusoid.

*//In this kind of fine-tuning, there should only be one learning epoch where shuffle=False, but shuffle=True can be imitated by "**rotation**" of the learning phases:*

EPOCHS = 1 # The number of times, the language model runs through the examples from the dataset.

Repetition is mother of knowledge when it occurs during learning (in the dataset) and father of dulling when it occurs during understanding/understanding (with more epochs of fine-tuning).

Usually, one learning epoch is not enough in this configuration. At the same time, no more than one epoch can be used, because this will lead to an overload of the same learning parameters.

//Explanation:

When performing fine-tuning through quantized low-rank adaptation, training is performed on only a small percentage of the total number of parameters of the language model. In this case, the dataset has a certain sequence:

(a = examples of learning, b = examples of understanding, and c = examples of work).

When you train the language model on a dataset without using shuffle (shuffle = True), if there is more than one learning epoch, the examples from the dataset will change the weights of the trained parameters in the same way, and this does not reflect well on the performance of the language model (degrades it).

The solution in this case is to use a rotational age of learning (3 in 1), which is equal to three epochs, and this is done as follows:

The dataset is created in such a way that the examples are in the following order:

- Epoch of learning 1: a, b, c (learn, understand, work);
- Epoch of learning 2: b, c, a (understand, working, learn);
- Epoch of learning 3: c, a, b (work, learn, understand);

//This is done through the connection script in Midextral.

//Only in the above way, the examples pass through the three groups of trained parameters of the language model, overlapping minimally, due to the different ratio of examples (in terms of number and percentage).

- Work after completing the instructional fine-tuning of the language model:

1. After completing the instruction fine-tuning, it is necessary to merge the LoRA adapters with the basic language model on which the training was conducted. In this way, the final trained model is obtained.
2. After the connection, it is necessary to convert the language model from .safetensors format to .gguf format, which is optimized for local execution and more efficient operation.
3. After conversion, it is necessary to quantize the language model in order to reduce the size, optimize performance and lower resource consumption.
4. After quantizing, the language model can be used and tested, and for this purpose it is necessary to make additional settings to improve the quality of the responses when generating.

III. Olachra - System for use of large language models - "work/action"

Only software such as "Ollama" can be used to use the training model. In this case, however, the language model works in a specially developed system, which is a combination of "Ollama", "LangChain" and "ChromaDB" (Olachra engine).

The name of the system in Bulgarian is "Олахра", and in English it is "Olachra" from Ollama, LangChain, ChromaDB.

The name contains the words (Ola and Ra). Ola colloquially means "Hello" in Spanish. Ra symbolically means "light/enlightenment". That is, the name "Olachra" literally means: "Hello light/enlightenment", which is analogous to the well-known: "Hello world" in programming.

//Literally translated from Spanish, Ola means "wave" and Hola means "Hello".

Background:

-Creating ability for learning - knowledge memory: By combining the programming language "Python" and the library "ChromaDB" with "Ollama", the declarative memory of the system is created. Through it, the language model can store knowledge and context (meaning), learn and use the experience gained over time. meaning), and not by keywords from headings, text and tables, because this is a vector database.

- Creating ability for understanding - skill memory: By combining the programming language "Python" and the "LangChain" library with "Ollama", the procedural memory (logic) of the system is created. The library allows the language model to structure and organize information, to plan, to make decisions (not just to draw conclusions), depending on the mode of operation.

- Creation of an information system for the working of the language model: The components of the system can be described figuratively as follows: "Ollama" - the body of the system, "LangChain" - procedural memory, "ChromaDB" - declarative memory, "LLM" (language model) - the brain.

Simplified principle of system operation in manual mode:

1. A user enters a prompt into the system, after which it is accepted by LangChain;
2. LangChain transmits a request to (ask) ChromaDB for information if it deems it necessary, using a language model for vectorising the data;
3. "ChromaDB" accepts the request and transmits (returns) several texts (information) in vectorized form;
4. "LangChain" receives the information and combines these texts (information), then transmits it to (sends to) "Ollama" as a final prompt;
5. Ollama receives the information and then generates a response based on it through the language model;

Principle of operation of the components of the system:

1. "ChromaDB" is the "declarative memory" of the system: "ChromaDB" is a vector database in which the "meaning" of the text (information) is stored, not just words. It converts text into a vector (numbers that represent the meaning), can be searched by meaning rather than keywords and given to the "declarative memory" model.

//**Declarative memory is the memory of knowledge.** Example: "I know what a given thing is."

2. "LangChain" is the "procedural memory" of the system: "LangChain" is a programming tool (framework – basis for application development and mediator) that connects the language model with memory, data and external tools. It can be used to create structures such as:

-Chain (chain) - a sequence of steps (example: "read text → summarize → translate").

-Memory - to remember previous conversations and add them to a new context.

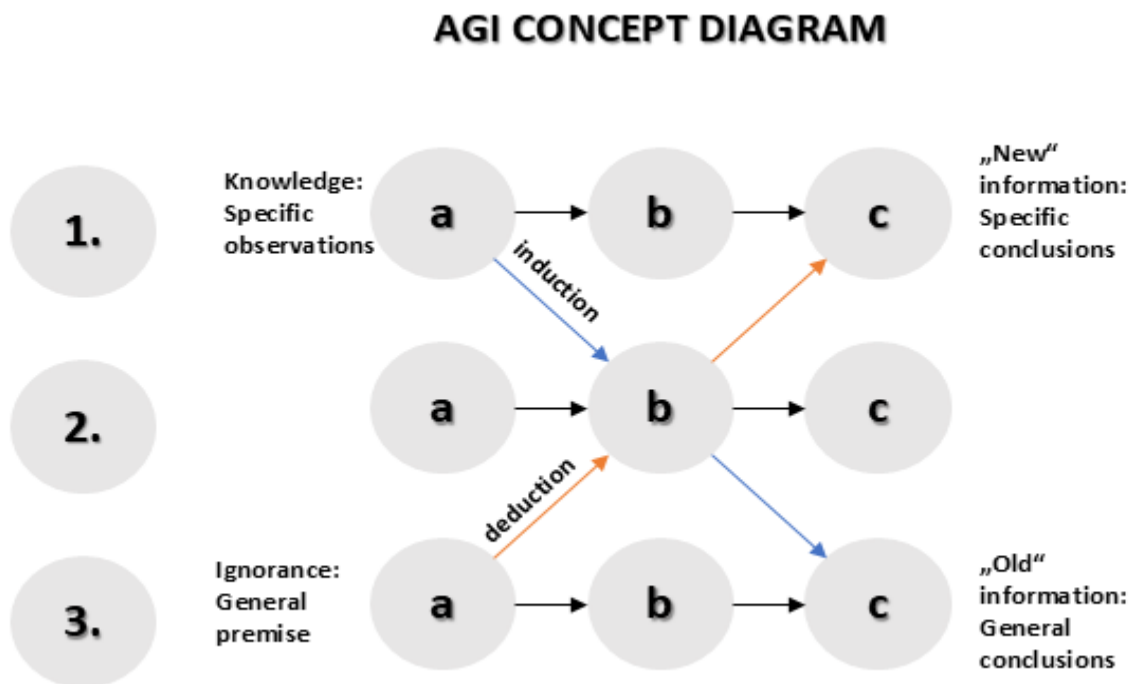
-Agent - allows the model to decide for itself what action to take.

Allows you to combine models, files, databases, APIs and other sources of information.

//**Procedural memory is a memory of skills.** Example: "I know how to do something."

3. **"Ollama" is the "body" of the system:** "Ollama" is a program that allows the use of language models locally on a computer - without the Internet, without external servers, etc. It performs the calculations of the language model (in the CPU or in the video card - GPU). It accepts your prompt/question (prompt) and returns an answer (completion). It works through a local API, which is why requests can be sent via "Python".

//Example: "I can do something."



Phases:

a = learn/knowledge;

b = understand/skill;

Understanding the information learned during fine-tuning is key because, as you can see in the diagram above, understanding is the intersection between induction and deduction.

c = work/activity;

"A focus on knowledge leads to development (induction), and a focus on ignorance leads to progress (deduction)." Familiar (old) information can be deduced from knowledge, and unknown (new) information can be deduced from ignorance. Focus on specific observations leads to general conclusions (development), and focus on general premise leads to specific conclusions (progress).

- By the method of induction (narrow artificial intelligence), general conclusions are drawn from the language model from specific observations. In induction, the focus is on knowledge, which is why the model becomes more rigid. In this case, the model typically can generate information that was present in the dataset during the training.

- By the method of deduction (artificial general intelligence), specific conclusions are drawn from general premise by the language model. In deduction, the focus is on ignorance, which makes the model more adaptive. In this case, the model typically can generate information that was not present in the data set during training.

//It can be said that ordinary artificial intelligence is any information system based on induction, and general artificial intelligence is any information system based on deduction that can combine both.

//In the "Magi" operating system, there is text under the logo: "powered by Olachra", because it means that the combination of Ollama + Langchain + ChromaDB is used, which in this operating system is like the holy trinity, also in the backend there is a "class OlachraEngine", which is of key importance.

//The system 4-4-2 is observed at the micro and macro level throughout the ecosystem. At micro level in the interactions inside each individual system (platform) and at macro level in interactions outside each system. At micro level in internal interaction in each individual system. At macro level in external interaction between systems.